

To Kill Three Stones with Six Birds: A Common Grammar for the SM, QM, and GR

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Version 1: 17 June 2026 · Version 3: 3 October 2026

Abstract

Can one small mathematical vocabulary organize two very different open problems in fundamental physics? We test this with the Six Birds emergence calculus, whose working core is simple: a coarse description is a quotient of a finer one, a quantity either passes through the quotient or leaves a checkable obstruction, and two descriptions may share a common refinement. We apply this vocabulary to two problems: why the Standard Model (SM) has its gauge structure, and how quantum (QM) and gravitational (GR) descriptions relate. Every result lives on an explicitly declared finite model (a finite enumeration, a fixed finite graph, or matrices of fixed size), holds only under the stated caps and conventions, and ships with a public validator.

On the SM side, a census of 1,066 chiral gauge structures shows that, within the declared windows, only the $SU(2) \times SU(3)$ family admits a “clean” breaking branch, and every clean branch breaks the $SU(2)$ factor. Selection is therefore a property of a structure-plus-branch pair, not of a bare gauge group. A separate toy theorem rules out single-factor $SU(N)$ candidates. An exact construction of the regular $SU(5)$ embedding forces the hypercharge direction and reproduces $\sin^2 \theta_W = 3/8$ and $k_Y = 5/3$. These are standard values. They are recovered under named conventions and do not single out $SU(5)$.

On the QM–GR side, an exact theorem shows that the declared QM and GR readouts of an abstract carrier are incomparable: neither is a coarse-graining of the other, although a common refinement carries both. On finite graphs, an exact max-flow/min-cut certificate realizes $\text{Area}(\min \text{ cut}) = \sum_e c_e y_e$ with per-edge shadow prices y_e , and sampled tensor-network entropies stay below the cut. In a toy holographic carrier, a general linear-algebra theorem shows that the full raw boundary readout loses only internal gauge data, while an exact witness shows that a half-boundary readout loses physical information.

Three questions are developed as construction programs. For the first, nineteen exact equal-cut graph pairs cannot be joined by forward reduction moves, while six state-level pairs turn out to be gauge-equivalent. The other two concern gravitational record formation and the persistence of records; their finite precursors are, respectively, conditional on an assumed channel and sensitive to how records are defined. Negative results, such as the exact commutation of the natural “quantize” and “curve” completions, are reported as results. No claim about nature is made beyond these finite carriers.

Keywords. Six Birds Theory; emergence calculus; finite structural constructions; Standard Model gauge selection; quantum gravity; holographic entanglement.

1 Introduction

Two of the deepest open problems in physics look nothing alike. The first is a *selection* problem: why the Standard Model has the gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$, its particular representations, three generations, and its scales [1]. The second is a *compatibility* problem: general relativity is

perturbatively non-renormalizable [2, 3], and no quantum theory of gravity has been confirmed experimentally. This paper asks a modest question about the two: can both be studied with one small, finite vocabulary, and if so, what does that vocabulary actually deliver?

The vocabulary comes from the Six Birds emergence calculus (Section 2). Its working core fits in a sentence. A coarse description is a *quotient* of a finer one; a quantity of interest either *descends* through the quotient or leaves a set of witnesses showing that it does not; and two coarse descriptions may share a *common refinement* that carries both. These are elementary notions. What makes them useful here is that they can be evaluated exactly on finite carriers, so each step can be checked, controlled, and, where it fails, recorded as a result.

A shared vocabulary is not a solution to either problem. It earns evidential weight only where it produces explicit constructions that could have come out differently and that survive review. For that reason the paper is built around finite, declared carriers, and every result carries a grade that states what kind of result it is. We use the calculus’s finite toolkit (obstructions, quotients, construction audit, and claim bookkeeping). We do not use its strict-extension, reclosure, promotion, endogenous-repair, or run-level dynamical machinery, and we supply no continuum bridge to a physical substrate.

Main results

1. **A branch-level selection of $SU(2) \times SU(3)$** (Section 4). A conjugation-consistent census contains 1,066 labelled genuinely chiral gauge structures (419 orbits under the declared symmetries, or 195 under an alternative charge normalization). Within the declared representation, charge, field, and scalar caps, only structures with nonabelian factors of dimensions 2 and 3 (the 2|3 family) admit a *clean* breaking branch. Every clean branch breaks the $SU(2)$ factor, and every $SU(4)$ branch is breaking. Every 2|3 structure also has a breaking branch, so the selection is of a structure-plus-branch pair, not of a bare gauge structure. A separate witness-independent toy theorem excludes stable single-factor $SU(N)$ candidates.
2. **Exact unification objects, recovered rather than predicted** (Section 4.4). For the regular $SU(5)$ embedding, exact matrix algebra forces a one-dimensional hypercharge direction and gives $\sin^2 \theta_W = 3/8$ and $k_Y = 5/3$ for the $\mathbf{\bar{5}} + \mathbf{10}$ package. Generator action identifies the X/Y coset roles. The ratios do not select the parent group, since Pati–Salam also gives 3/8. Under a declared content quotient, the SM and its candidate alternative are one orbit.
3. **QM and GR readouts as siblings, and a finite min-cut duality** (Section 5). On an abstract Boolean carrier, the declared QM and GR coordinate readouts are mutually non-factorizing: neither is a coarse-graining of the other, though both factor through a common refinement. A 54-record field-plus-background construction realizes such readouts under stated conditions. On finite graphs, exact LP certificates give $\text{Area}(\text{min cut}) = \sum_e c_e y_e$, with the y_e the per-edge shadow prices. An exact computation also shows that the natural “quantize” and “curve” completions on this carrier *commute*, so this level offers no route-mismatch explanation of why quantizing gravity is hard.
4. **Three construction programs** (Section 6). The first asks whether the same min-cut data can hide different geometries. Its graph-level subproblem shows that, for nineteen exact equal-cut pairs, the two sets of presentations reachable by forward reduction moves are disjoint. Whether they are disjoint under arbitrary moves in both directions remains open. Its state-level subproblem finds six exact pairs with identical cut and state data, all of which turn out to be gauge-equivalent tensor networks. The second and third programs, on whether gravity must act as a classical record channel and on which invariant operators persist as records, remain open. Their finite

Standard-Model track Section 4	QM–GR track Section 5	Programs & applications Sections 6–7
Gauge census. 1,066 chiral structures; 52 survive three filters (419 $\rightarrow$ 24, 195 $\rightarrow$ 24 as orbits).	Route mismatch. The quantize and curve completions commute exactly.	P1, graphs. 19 equal-cut pairs have disjoint forward-reachable sets.
Branch selection. Only 2 3 admits a clean branch; every clean branch is $SU(2)$ -active.	Access fork. QM and GR readouts are incomparable; a common refinement carries both.	P1, states. Six equal-data pairs are gauge-equivalent (general lemma).
Single-factor exclusion. No stable single-factor $SU(N)$ is clean (toy theorem).	Field provenance. 54 field-plus-background records realize the fork under stated conditions.	P2, BMV. An assumed perfect-record channel is LOCC, hence BMV-null.
$SU(5)$ frame. Hypercharge forced; $\sin^2 \theta_W = 3/8$, $k_Y = 5/3$ under named conventions.	Min-cut duality. Area = $\text{OPT}_{\text{dual}} = \sum_e c_e y_e$ with exact certificates.	Open. Symmetric-closure separation (P1); deriving the record channel (P2); record persistence (P3).
Content and masses. One content orbit in the template; charged ranks $14N$, $N \leq 4$.	Tensor-network entropies. $S \leq \text{min cut}$; mean $S/\text{cut} = 0.73, 0.90, 0.94$ at $D = 2, 3, 4$.	Information loss. Full raw boundary loses only gauge data; half boundary loses physical data.
Generations, scales. No N -blindness, no selection of $N = 3$; naturalness surface spans 0–40.6%.	Linear response; one ledger. No Einstein match at a nondegenerate point; Born fails min-plus gluing.	Cosmological constant. Fixed points for every sampled background; no in-layer value law.
theorem (written proof) exact finite computation conditional construction or recognition sampled numerics negative result open program		

Figure 1. The results at a glance. Each box is one statement, colored by the kind of evidence that supports it. “Exact finite computation” means an exhaustive or exact-arithmetic computation on a declared finite carrier; “theorem” means a written proof whose hypotheses are stated in the text. Every statement holds only on its declared carrier and under its stated caps and conventions; none is a claim about nature.

precursors are, respectively, conditional on an assumed channel and sensitive to how records are defined.

- Further applications and two toy theorems** (Sections 4, 5, and 7). The single-factor exclusion and the coordinate-partition incomparability are genuine theorems on their toy carriers. In a toy holographic carrier, a general factorization theorem shows that the full boundary readout loses only gauge data, while a half-boundary readout provably loses physical information. Bell, cosmological-constant, and measurement rows carry their own finite grades. None establishes universality.

Figure 1 places these results side by side, colored by the kind of evidence behind each one.

Scope. Nothing here derives Standard-Model parameters, solves quantum gravity, proves a universal composition law, or transfers a statement from these carriers to nature. Two words are used in a narrow sense throughout. A *recognition* means that a known exact structure is reproduced once all named inputs are declared. An *open program* means that the missing construction has been identified and the repository isolates exactly what would have to be built.

How to read the paper. Section 2 introduces the vocabulary with a worked picture, and Section 3 describes how results were checked. Sections 4 and 5 present the two tracks, Section 6 the three construction programs, and Section 7 four further applications. Section 8 asks what

the two tracks share, and Section 9 collects every statement with its grade and its most direct test. The appendices give formal definitions (A), reproducibility details (B), the record of rejected constructions (C), notation (D), and the version history (E).

2 The Six Birds emergence calculus, in brief

This section is self-contained and assumes no prior exposure to the framework. The formal development is in the foundations papers (Foundations II–IV [4, 5, 6]; *Foundations of Emergence Calculus* [7]; *Why Mathematics Even Works* [8]), and Appendix A states every formal object used here. What follows is the minimum needed to read the rest of the paper.

2.1 Layers, quotients, and descent

Physical description is layered [9]: thermodynamics sits over statistical mechanics, hadrons over quarks, classical fields over quantum states. The calculus models a layer as a *quotient* of a finer description. Three ingredients are needed:

- a **carrier** H , the set of finer-grained configurations (microstates, field configurations, tensor data);
- an **access map** $q : H \rightarrow Q$, a surjection that forgets what the layer cannot see, so that two configurations look identical to the layer exactly when q sends them to the same point;
- a **readout** $t : H \rightarrow T$, a quantity one would like to evaluate.

The question asked again and again in this paper is whether the readout **descends** to the layer, that is, whether the layer can compute it:

$$\begin{aligned} t \text{ descends through } q &\iff q(h) = q(h') \text{ implies } t(h) = t(h') \text{ for all } h, h' \in H \\ &\iff t = \bar{t} \circ q \text{ for some } \bar{t} : Q \rightarrow T. \end{aligned}$$

When descent fails, the failure is recorded as an **obstruction set**

$$\mathcal{O}_q(t) = \{(h, h') : q(h) = q(h') \text{ and } t(h) \neq t(h')\}.$$

Each pair in it is a *witness*: two configurations the layer cannot tell apart that nonetheless differ in t . Figure 2 shows the idea on a toy carrier. When the carrier is a finite set, these sets can be enumerated exactly, and many of the constructions below reduce to showing that a particular obstruction set is empty or not. Where the configuration space is not finite (real capacities on a fixed graph, or real matrices of fixed size), the corresponding statements are proved as general theorems and only their finite instances are checked by computation.

The same test compares two maps on one carrier. For π_0, π_1 defined on S , the **factorization defect** is

$$\Delta_{\text{fact}}(\pi_0, \pi_1) = \{(s, s') : \pi_0(s) = \pi_0(s') \text{ and } \pi_1(s) \neq \pi_1(s')\},$$

and $\Delta_{\text{fact}} \neq \emptyset$ exactly when π_1 is *not* a function of π_0 . This elementary equivalence (Appendix A.2) does a great deal of work. In Section 4 it sorts symmetry-breaking branches into clean and breaking ones; in Section 5 it shows that a gravitational readout is not a coarse-graining of a quantum one. Using the same test in both places does not identify the physical objects involved.

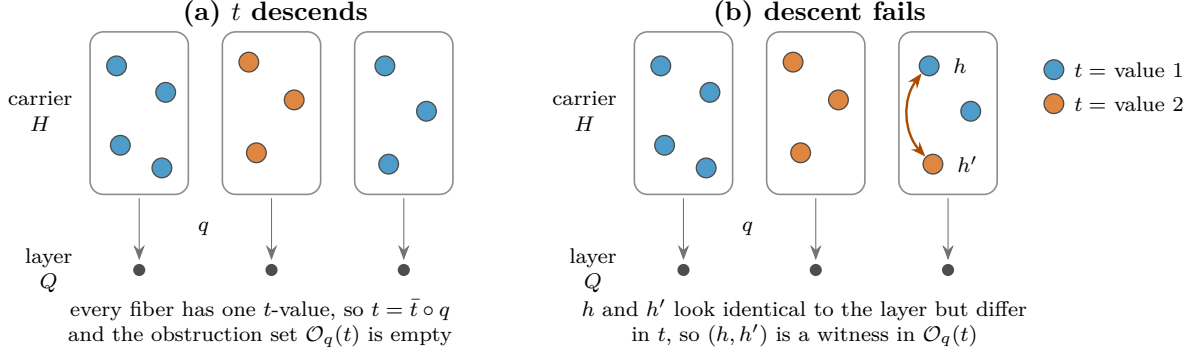


Figure 2. Descent and obstruction on a toy carrier. Boxes are the fibers of the access map q : sets of configurations that the coarse layer cannot distinguish. Colors are values of a readout t . (a) Every fiber is one color, so the layer can compute t . (b) One fiber contains two colors; the pair (h, h') is a witness in the obstruction set. The factorization defect $\Delta_{\text{fact}}(\pi_0, \pi_1)$ is the same construction with $q = \pi_0$ and $t = \pi_1$.

2.2 Six role names

The calculus files its judgments under six **primitive roles** $\mathbb{P} = \{P_1, \dots, P_6\}$, which name the recurring ways in which layers relate (Foundations III):

role	name	what it asks, and where it appears here
P_1	descent	does a property pass through a quotient? (every descent and obstruction test)
P_2	representability	what can a layer admissibly express or select? (the SM selection layer, Section 4)
P_3	route mismatch	do two routes to “the same” object fail to commute? (the quantize/curve pair, Section 5.1)
P_4	refinement	does one layer refine another? (common refinements; the $SU(5)$ frame and the QM-GR parent)
P_5	packaging	is structure closed into an auditable unit? (closure chains and record packets)
P_6	audit	was anything smuggled between layers? (the checks of Section 2.4)

The six are role names, not an algebra, and the calculus says so formally. Foundations III proves that no total binary operation on the six labels can decode the status of a typed judgment: it exhibits two judgments with the same pair of labels but different statuses, so status is not a function of the pair. (This is weaker than saying that no binary operation on six symbols exists, which would be false.) Nor does the calculus claim that all of physics decomposes into six primitives.

On top of the roles sits a catalog of **structural laws** (“F-laws”, Foundations IV [6]): statements about quotients and closures that hold on any substrate satisfying their hypotheses. Each law used here has a one-line form in Appendix A.5, together with the inputs it needs. Briefly: F23 (probability as a stable measure on an unresolved fiber, given that measure); F26/F47 (contingency and fine-tuning); F27 (conservation as descent through a supplied orbit relation); F34 (information loss as a nonempty obstruction set); F37 (complementarity as the absence of a joint quotient in a declared admissible class); F39 (entropy as fiber volume, under its stated chain-rule premises); F48 (topological obstruction as a nontrivial gluing class); F49 (common-source factorization of correlations); F50 (background selection: necessary, moduli, or vacuum); F51 (unification as common

refinement). A law supplies a form, not its premises: wherever one is used, the measure, orbit relation, admissible class, or gluing data it needs is either constructed or named as an input.

2.3 What counts as a result: grades

Because the calculus is built to audit claims, every result carries a grade (a *landing mode*), and the grade is part of the result:

- **COMPUTE**: a number, relation, or verdict drops out of the construction.
- **SELECT**: the construction picks a structure out of a declared space, on declared conditions.
- **RECOGNITION**: a known result is reproduced as an instance of a law, without the target being used as an input (for example the min-cut/LP identity of Section 5.4).
- **FORBIDDEN RULE / DEMARCATION**: a configuration is excluded, or a quantity is classified as law-fixable versus contingent, with a control on which the verdict comes out the other way.
- **PROVE-BLIND** and **GROUND** complete the vocabulary but are not used: no result here proves that a quantity is undeterminable on a declared class, and no condition is supplied as the shadow of a named higher source. The two natural candidates for GROUND are stated as hypotheses or open programs (Sections 4.5 and 5.3).

The grades keep three distinctions explicit: a selection is not a derivation, a recognition is not a discovery, and a forbidden rule on a toy carrier is not a fact about nature. Attaching a grade cannot rescue a claim whose carrier, quantifier, or bridge to physics has not been built.

2.4 Guarding against smuggled conclusions

Any flexible framework risks building its conclusion into its premises. The audit role P_6 is a set of checks against this, mechanical where they are implemented. A quantity may not be *defined* to equal its target (anti-tautology). Each hypothesis must be satisfiable by some structure that violates the conclusion (anti-circularity). Imported machinery is frozen and pinned by hash, so it cannot be retuned toward an answer. Every positive claim needs a **can-fail control**, a configuration on which the claim comes out false, which shows that the test actually discriminates. Section 3 describes how these checks operated in practice, including the many cases in which they rejected our own constructions.

3 Method: finite audited carriers

The paper combines exact finite computations with a smaller number of bounded numerical probes (fixed-point solvers and sampled tensor networks). The protocol is designed to expose circularity and dependence on arbitrary choices of representation. It cannot turn a finite carrier into evidence about nature, and it does not try to. Figure 3 summarizes the workflow; Appendix C records the constructions it rejected.

3.1 Declared, frozen carriers

Each construction begins by declaring a finite carrier: the gauge-structure census of Section 4, or the graphs, tensor networks, and mode grammars of Section 5. Obstruction sets, factorization defects, entropies, min-cuts, and kernels are then computed deterministically on it. Carriers, predicates,

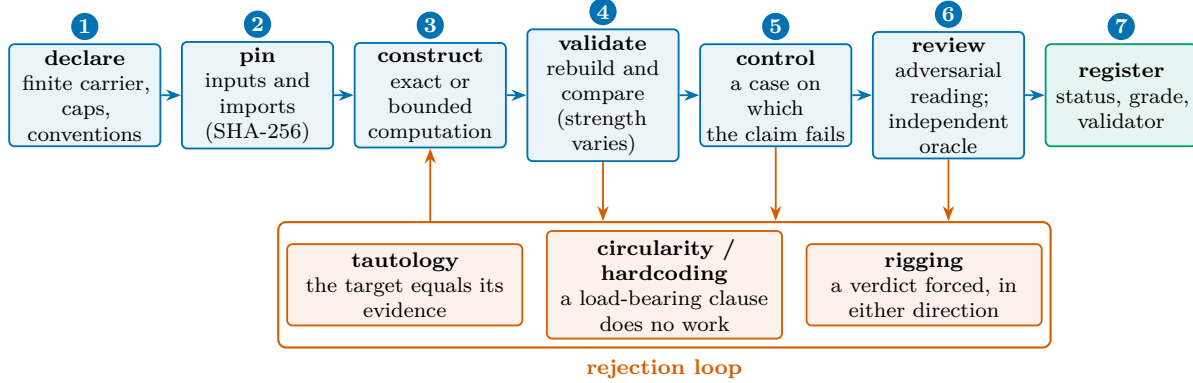


Figure 3. How each result was produced and checked. Steps 4–6 can send a construction back to step 3. The three defect classes caught this way are described in Section 3.3 and Appendix C. Validator strength varies, from full byte-level rebuilds to checks of stored summaries (Appendix B); review is internal and model-assisted, not journal peer review.

and imported code are pinned by SHA-256 hashes so that they cannot be retuned after the fact; Appendix B lists where this pinning is incomplete.

Every construction has a validator, but validators differ in strength. The strongest rebuild the entire result in memory and compare every output byte for byte. Some older ones check stored summaries instead of recomputing them; Appendix B lists exactly which. A passing validator is therefore necessary but never sufficient. The audit also read the constructions for meaning, regenerated carriers, and added can-fail and mutation controls.

Two further kinds of check do not depend on the production code. An independent representation-theory oracle enumerates semistandard tableaux and evaluates weight characters with the Weyl denominator, instead of reusing the production Littlewood–Richardson routine. It reproduces the dimensions, conjugates, Dynkin indices, and cubic anomalies of all fourteen canonical representations, all 368 ordered triple singlet counts, and every one of the 65 distinct singlet queries made by the record census of Section 6.3. Tampering controls confirm that the strengthened validators reject forged artifacts even when their summary counts or verdicts are left unchanged.

3.2 Can-fail controls

A claim that cannot come out false is not a result. Each principal positive computation is paired with a control:

- the SM selection’s “collapse” verdict is paired with a broad-measure landscape control that must *not* collapse, and with 204 refinement orders, none of which flips the verdict;
- the directed QM–GR non-factorization is paired with a *nested* readout for which factorization must, and does, succeed;
- the holographic monogamy inequalities are paired with a GHZ state that must, and does, violate them;
- the P1 graph program is paired with exact active-cut quotient regressions;
- the information-loss theorem is paired with a rank-deficient pair that has the same boundary state but is not gauge-related, showing that its full-rank hypothesis is needed;
- the record-stability census must count a scalar-dressed regression operator and must exclude a spurious spectator-Cartan token.

3.3 Self-correction in both directions

Over the course of the program the checks rejected our own constructions many times. The defects fell into three classes (full accounts in Appendix C):

1. **Tautologies**, where a “coincidence” is an identity. An early area–entropy “match” computed the von Neumann entropy from the Schmidt spectrum on one side and the Shannon entropy of the squared singular values on the other: the same number twice, agreeing to machine precision. The accepted version (Section 5.4) uses the geometric min-cut, which does not depend on the state: it stays fixed under reseeding while the entropy moves.
2. **Circularity and hardcoding**, where a load-bearing clause does no work, for example a compatibility flag that was constantly true. The surviving evidence in Section 5.4 computes mutual-information inequalities on contracted states and runs the same pipeline on a GHZ control.
3. **Rigging toward a verdict, in either direction.** The entanglement–geometry program (Section 6.1) needed several attempts. One was rigged toward the positive verdict (a hand-built graph with the “invisible mode” built in). One was rigged toward the negative verdict (a dimension hardcoded to zero, plus a scoring filter chosen to dismiss the answer). One was inflated by a degeneracy: a one-sided finite-difference Jacobian at a maximally tied point, betrayed by residuals that scaled linearly with step size. All three were rejected. An exact quotient analysis then showed that every null direction of the naive probe was a reparametrization, which is what led to the exact program of Section 6.1.

We call this discipline **symmetric scrutiny**: deflationary conclusions are checked as hard as positive ones. Appendix C keeps the rejected attempts on record rather than presenting the final constructions as if they had been there from the start.

3.4 Review and separation of the two tracks

Every load-bearing claim went through an adversarial review workflow. One agent reproduced each review finding, a separate read-only reviewer assessed the constructions, and changes were committed only through a gate. The outcome is a claims registry, `review_2026/CLAIMS_REGISTRY.md`, which records for each claim its status, evidence type, controlling source, and validator, and which this paper follows at each claim site. A separate mathematics review re-examined all 42 registry records. This is an extensive internal and model-assisted audit record. It is not journal peer review or independent replication.

A comparison between the tracks is meaningful only if neither construction was copied from the other. The program therefore ran under an **anti-contamination guard**. The two substrates were typed differently from the outset (a selection layer for the SM, a common refinement for QM–GR). SM-track validators fail if field-carrier tokens from the other track appear, and steps that pattern-matched one substrate onto the other were rejected. This guard is procedural self-report, not independent blinding, and it does not establish universality.

4 Result I: a finite Standard-Model selection construction

This section treats the Standard Model’s gauge structure as something *selected* from a finite space of alternatives and asks what the selection depends on. The short answer is that, within declared windows, a “clean” symmetry-breaking condition singles out the $SU(2) \times SU(3)$ family, but only as

a property of a structure together with a breaking branch. Figure 4 summarizes the census and the branch typing.

4.1 The selection layer

The construction (called L^*) places a finite space of candidate structures, a set of constraints, and a selector one layer above the realized candidate. Its descent and role-split diagnostics form a coherent finite construction, not a derivation from nature. Two caveats are essential. First, the construction assumes a single selection layer; the supporting evidence shows block structure only within that layer and does not exclude hidden or multi-layer selectors. Second, the *architecture* of the selector is not determined. Two constructed families, a scale-carrying memory layer and a finite naturalness budget, both form, are admissible, and reach a closure fixed point on the 468-row admissible carrier. The memory-layer closure gate is an experimental finite diagnostic, not evidence of a physically independent scale layer (review_2026/repairs/s5_f24_f47/).

4.2 The carrier and what it excludes

The carrier enumerates candidate chiral gauge structures within declared windows: one or two nonabelian factors with $SU(N)$ dimensions 2 through 4, a field cap of five, a component cap of six, and a declared $U(1)$ charge alphabet. Representation conjugation is implemented as an involution, including the special cases of pseudoreal and real $SU(2)$ representations, so the alphabet is closed under conjugation and contains no aliases. This matters: an alphabet without this closure inflates the count to 11,990 rows, of which 9,126 are duplicates. Dimensions, Dynkin indices, and cubic anomalies are checked against an independent table, and again by the independent oracle of Section 3.1.

Within these windows the carrier contains 1,066 labelled, genuinely chiral structures. How many of them are “different” depends on a declared equivalence. Freely adjoined, fully neutral singlets are deleted, and the orbit convention identifies structures related by exchanging equal-dimension factors, conjugating all nonabelian representations at once, or flipping the sign of the $U(1)$ charge. It does not identify charge rescalings or field-by-field conjugation. This gives 419 orbits. Normalizing charges to primitive integers instead gives 195 orbits; this is an alternative convention, not a convention-free physical count. Three successive filters, atomic packaging, closure consistency, and chirality-faithfulness, then leave

$$1,066 \rightarrow 84 \rightarrow 62 \rightarrow 52, \quad 419 \rightarrow 37 \rightarrow 27 \rightarrow 24, \quad 195 \rightarrow 37 \rightarrow 27 \rightarrow 24,$$

a cumulative exclusion of 95.12%, 94.27%, and 87.69% respectively. The 52 survivors are the chirality-faithful structures used below. These percentages belong to their exact denominators and are not a measure on gauge theories in general. The representation model, quotient mutations, and an independent scalar gate are in review_2026/repairs/s1_carrier_reconstruction/.

4.3 Clean separation selects a branch, not a structure

For each chirality-faithful structure C , the construction enumerates *every* admissible complex scalar ϕ in the inherited alphabet and cap, rather than picking one witness, and separately every scalar pair (ϕ_1, ϕ_2) that jointly covers the Yukawa couplings and jointly breaks. Each such choice is a *branch*. On each branch the construction applies a fixed residual proxy, in which the factor $SU(N)$ on which the scalar acts breaks to $SU(N-1)$, and computes the active factor, the residual structure,

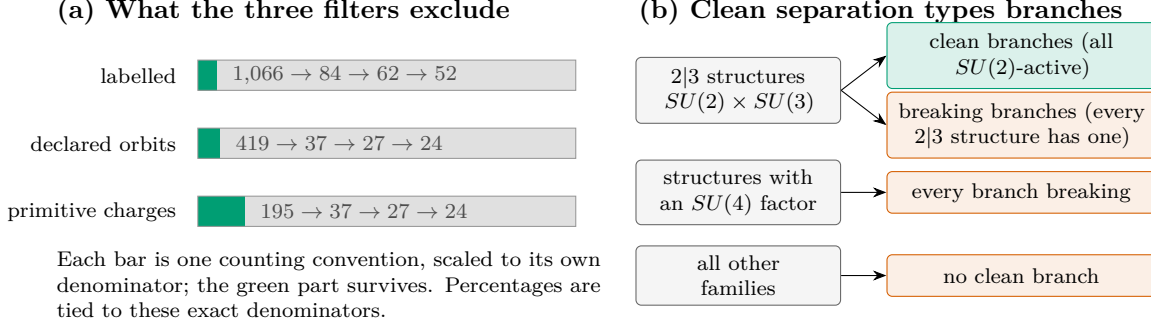


Figure 4. The Standard-Model census. (a) Within the declared windows, three successive filters (atomic packaging, closure consistency, chirality-faithfulness) remove most candidate structures under each of three counting conventions. (b) A branch is a structure together with a choice of breaking scalar (or scalar pair). Clean branches exist only for 2|3 and always act on the $SU(2)$ factor, but every 2|3 structure also has breaking branches. The selection is therefore existential over branches, not a property of the bare gauge structure.

the number of broken (coset) generators, and the factorization defect between the confinement readout and the mass readout. A branch is called *clean* when this defect is empty:

$$\text{clean separation} \iff \Delta_{\text{fact}} = \emptyset.$$

Both cleanliness and the defect are functions of the same proxy count, so the census is a classification conditional on the proxy, not a derivation of symmetry breaking.

Only 8 of the 52 structures have an admissible singleton branch (12 branches in all); every one of the 52 has admissible scalar-pair branches. Within the declared singleton and two-scalar caps, the result is as follows. Clean branches occur only in the 2|3 family, with nonabelian factors $SU(2)$ and $SU(3)$. The four labelled clean singleton structures form a single carrier orbit, and the 68 clean two-scalar rows form 17 pair-branch orbits. Every clean branch is $SU(2)$ -active, meaning that the scalar acts on the $SU(2)$ factor; every branch with an $SU(4)$ factor is breaking. The quantifier is what matters:

- “every branch is clean” is false for 2|3: every 2|3 structure also has a breaking branch, so clean separation is not a property of the bare structure;
- “some branch is clean” holds only for 2|3: clean separation selects the branch class (2|3, $SU(2)$ -active orientation).

No clean branch outside this class appears in the two-scalar extension either. The result is a uniqueness statement on an explicitly finite domain.

A toy theorem for one factor. Separately, a witness-independent theorem excludes single-factor candidates in its toy grammar. Suppose a single-factor $SU(N)$ structure passes the grammar’s stability condition, which requires a confining residual. The breaking scalar must then act on the only nonabelian factor, so the residual is $SU(N-1)$ with $N-1 \geq 2$, and the broken coset contributes $2(N-1) > 0$ massive vectors charged under the confining residual. The grammar’s deliberately coarse confinement readout gives these massive vectors the same value as the massless residual generators, while their mass status differs. That pair is a witness, so $\Delta_{\text{fact}} \neq \emptyset$ for every N . The validator rebuilds all released artifacts and constructs the actual witness pairs for $N = 3, \dots, 12$, with a same-mass control (which must remove the defect) and a clean two-factor control. This is a theorem about the toy grammar, not about confinement or the Standard Model in nature. A

companion exclusion of three active factors is weaker: it rests on the declared component cap, since a scalar charged under three nonabelian factors has at least $2^3 = 8 > 6$ components.

4.4 An explicit $SU(5)$ frame and the recovered ratios

Grand unification [10] is reconstructed here as an explicit object rather than assumed. The construction starts from exact 5×5 traceless Hermitian generators. For the regular embedding of $SU(3) \times SU(2)$, the commutant of the embedded algebra inside $\mathfrak{su}(5)$ is one-dimensional (the linear constraints have rank 23 on the 24-dimensional algebra). Tracelessness and commutation therefore force the hypercharge direction, up to scale and sign, to be proportional to $\text{diag}(-1/3, -1/3, -1/3, 1/2, 1/2)$. The twelve coset generators transform as $(\mathbf{3}, \mathbf{2})_{-5/6} \oplus (\bar{\mathbf{3}}, \mathbf{2})_{+5/6}$.

Decomposing the $\bar{\mathbf{5}} + \mathbf{10}$ package and adopting the named weak-pair normalization gives $\text{Tr } T_3^2 = 2$ and $\text{Tr } Q^2 = 16/3$, hence exactly

$$\sin^2 \theta_W = \frac{\text{Tr } T_3^2}{\text{Tr } Q^2} = \frac{3}{8}, \quad k_Y = \frac{5}{3}.$$

These are standard group-theoretic facts, reconstructed exactly with every input declared. They hold *given* the regular embedding, the fermion package, and the normalization convention. They do not select the parent group: the Pati–Salam group [11] also gives 3/8, and no well-posed alternative product parent with a different ratio has a constructed realization. They are not low-energy predictions either, since no running to a measured scale is performed. The contribution is an audited reconstruction, not a new derivation.

Inside the same frame, the roles of the coset are computed from generator action rather than assigned as labels. Generator action shows that the coset transitions carry $\Delta B \neq 0$. With $H = (SU(3) \times SU(2) \times U(1))/\mathbb{Z}_6$, the extracted cocharacter lattice, combined with three imported facts (H is connected, $\pi_1(SU(5)) = 0$, and $\pi_2(SU(5)) = 0$), gives the monopole class $\pi_2(SU(5)/H) \cong \mathbb{Z}$. So the proton-decay, monopole, and clean-separation roles all attach to one object, inside this constructed frame. A six-generator $SU(4) \rightarrow SU(3)$ object from the earlier toy grammar is a structural analogue only. It is not the twelve-generator X/Y coset, and a computed $SU(2)$ commutator shows that a fixed six-generator slice is not invariant (review_2026/repairs/s3_generator_construction/).

4.5 Does record stability force clean separation?

A natural hope is that clean separation is not an extra condition but follows from something deeper, such as the requirement that the low-energy theory support stable records. On this carrier the answer depends on what counts as a record. On the 12 admissible singleton branches, at the reference record threshold, a restricted record-token grammar makes the implication hold. A bounded but complete census of candidate invariant operators makes it fail in every reading, with or without scalar dressings. Section 6.3 gives the details. The phrase “gauge shadow of a memory layer” is therefore a narrative over a correlation, not a computed projection between layers, and the question of which candidate operators persist as physical records is an open program.

4.6 Content, generations, masses, and scales

Content. Under global color conjugation, global $U(1)$ sign, factor exchange, and deletion of inert singlets, the 28 labelled anomaly-free supports in the declared template form 14 orbits. The strongest conjunction of conditions selects one content orbit, and the SM and its superficially different alternative turn out to be two presentations of that orbit. This is uniqueness *within the template*; the associated minimality no-go has rank 14/14.

Generations. The grammar neither ignores the number of generations N nor selects $N = 3$. A materialized probe for $N = 1, \dots, 4$ finds the anomaly, parity, chirality, branch, and mass predicates unchanged by replication, while atomicity and route-incidence fail for every $N \geq 2$ for reasons internal to the grammar. Structural conditions that would make the grammar blind to N reduce to “at least one family”.

Mass-matrix rank. Rank is computed with independent symbolic coefficients and with multiplicities retained; mere coverage of the coupling graph does not determine rank. For the renormalizable one-scalar grammar, the generic charged ranks are 14, 28, 42, 56 for $N = 1, 2, 3, 4$, and the neutral deficiencies are 1, 2, 3, 4, one unpaired left-handed neutrino per generation. No masses, textures, or measured values are derived (review_2026/repairs/s4_content_quotient/).

Naturalness. The F47 naturalness fraction is a surface, not a number. In one declared cell (weighted admissible rows, upstream threshold) it is 2.247%, but that cell is not uniquely privileged: across the declared sweep the fraction ranges from 0 to 40.6%. The realized point sits exactly at the threshold 2.75, so a strict comparison excludes it.

4.7 Summary

question	status	scope
gauge-structure exclusion	finite census	1,066 $\rightarrow$ 52, 419 $\rightarrow$ 24, or 195 $\rightarrow$ 24 through three filters, under the named equivalences
clean-separation selection	branch-level finite selection	unique clean branch class: 2 3, $SU(2)$ -active; no selection of a bare structure
single-factor exclusion	toy theorem	witness-independent in the stated grammar; validator rebuilds artifacts and witnesses
hypercharge and ratios	conditional exact construction	regular $SU(5)$ embedding, $\bar{\mathbf{5}} + \mathbf{10}$, weak-pair normalization; imported topology for the monopole class
content	unique orbit in template	recognition within the template; minimality rank 14/14
generations	no blindness theorem	mixed N -dependence for $N = 1, \dots, 4$; no selection of three
record-stability grounding	open program	bounded candidate census is token-definition-sensitive
architecture and scale	underdetermined; sensitive	two architecture families close; F47 surface 0–40.6%

5 Result II: QM and GR readouts and finite graph duality

This section asks how a quantum description and a gravitational description of one system can relate, using the quotient vocabulary of Section 2. Three findings emerge. A much-discussed explanation, that “quantizing” and “curving” are two routes that fail to commute, has no support at this level: the natural finite versions commute exactly. Instead, the two readouts are *siblings*: neither is a coarse-graining of the other, but a common refinement carries both. Finally, on finite graphs, the familiar relation between minimal cuts and entanglement appears as an exact linear-programming duality. Figure 5 illustrates the second and third findings.

5.1 Route mismatch: the two completions commute

For two idempotent maps E_1, E_2 on a finite carrier, the set

$$\Delta_3^{\text{comp}}(E_1, E_2) = \{c : E_1 E_2(c) \neq E_2 E_1(c)\}$$

measures their failure to commute, and non-commuting examples exist (Appendix A.3). The natural QM–GR instance, however, returns a null verdict. On the 16-state Boolean carrier introduced below, the “quantize” and “curve” completions are coordinate conditional expectations: 16×16 averaging projections acting on functions (or distributions) over the uniformly weighted cube, not maps of its sixteen points. Both are idempotent, and they **commute exactly**: $E_{\text{QM}} E_{\text{GR}} = E_{\text{GR}} E_{\text{QM}}$, so no basis state is sent to different distributions by the two orders of composition, and the defect set is empty. The reason is that the overlap of the two coordinate sets factorizes on the Boolean cube. A naive normalized distance between the two encoded tables (0.5, or $1/\sqrt{2}$ after recoding 0/1 as ± 1) depends on the encoding and is not a commutator.

Non-commuting idempotent pairs can be built from non-factorizing partitions or constrained projections, and they serve as mathematical controls, but none of them arises from the frozen quantize and curve routes. At this level there is therefore no route-mismatch explanation of why quantizing gravity is hard. The exact matrices and recoding controls are in [review_2026/repairs/q2_route_mismatch/](#).

5.2 A fork, not a ladder

Consider an abstract carrier of four binary coordinates and three readouts:

$$q_{\text{QM}} = (d_0, d_1, d_2), \quad q_{\text{GR}} = (d_0, d_2, d_3), \quad L = (d_0, d_1, d_2, d_3).$$

If gravity were a coarse-graining of quantum mechanics (a “ladder”), q_{GR} would be a function of q_{QM} . It is not: in each direction there are eight unordered witness pairs (sixteen ordered pairs in Δ_{fact}), pairs of states that one readout identifies and the other separates. A nested control, in which one readout’s coordinates contain the other’s, factors as it should, and both readouts factor through L by construction. Exhausting all 16 coordinate-subset partitions gives 55 incomparable pairs among the 120 unordered pairs of distinct partitions. This is an exact toy theorem, conditional on the abstract carrier and on reading its coordinates as access. It is consistent with the commutation result of Section 5.1: commuting completions and incomparable readouts are different properties.

Where the coordinates come from. Bare coordinate labels say nothing about fields, so a separate finite construction supplies them. On 54 constructed records, each a field ψ together with a background potential V_0 , coarse field functionals and explicit phase-only and potential-only controls realize mutually non-factorizing QM and GR readouts. The conditions are material. The witness for the forward direction changes V_0 ; the QM readout includes the probability current as well as the Born density; d_0 is constant; and although the coordinate alphabets have sizes 1, 2, 8, 3, the 54 records occupy only 15 joint coordinate values. These 54 records and the abstract carrier are distinct objects. Nine hand-instantiated alternative architectures were tested; two reconcile the readouts, but both are partition-equivalent to L . This is **conditional finite provenance**: it shows that the fork can be realized under explicit choices, not that the family, type, or instance is unique ([review_2026/repairs/q1_access_provenance/](#)).

A related statement, labelled $T_{\text{QG-NoGo}}$ in the repository, tests two special “fused” or “stapled” routes. It does not quantify over general fused objects, coherent mediators, or operator algebras, and it cannot support any statement that quantum gravity is impossible within the grammar.

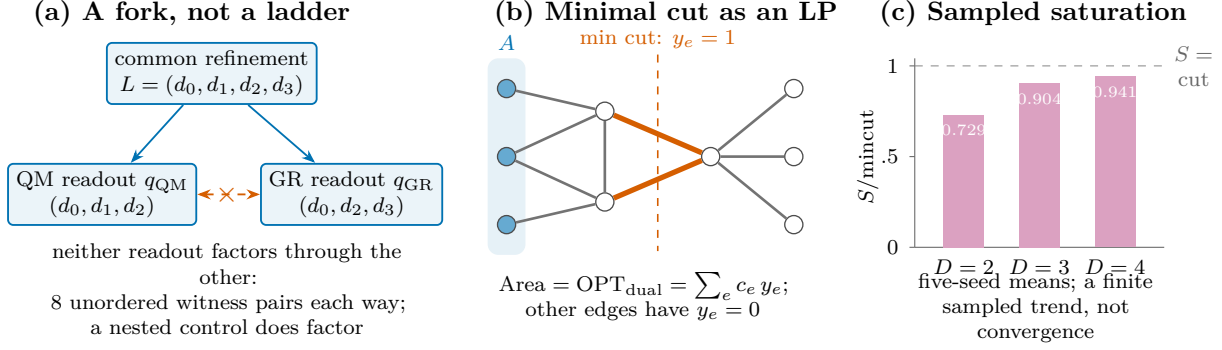


Figure 5. The QM–GR constructions. (a) On the abstract Boolean carrier, the QM and GR readouts are incomparable, yet both factor through the common refinement L . (b) A schematic boundary-anchored network: the minimum cut separating region A from the rest has value equal to the dual optimum, and the cut-incidence variables y_e are the per-edge shadow prices. (c) Entropies of independently contracted random tensor states, as a fraction of the min cut, at bond dimensions $D = 2, 3, 4$: means over five seeds on the sampled carrier.

5.3 The common carrier is a hypothesis

Nothing above shows that nature has a common carrier for the two readouts. Semiclassical co-sourcing, in which one field ψ supplies both a Born-type readout and the stress-energy that sources a potential, is a plausible physical motivation. But the available controls test different predicates, and the argument linking them is prose rather than a computation. The premise is therefore carried as a declared hypothesis, not as a grounded result. The 54-record construction shows that a finite realization can be built under explicit choices; it does not show that nature, the free gravitational sector, or a deep quantum-gravity regime shares such a carrier.

5.4 Minimal cuts as an exact linear program

In holography, the Ryu–Takayanagi formula relates the entanglement entropy of a boundary region to the area of a minimal surface [12, 13]. Its simplest discrete analogue replaces the surface by a minimum cut in a weighted graph, and the max-flow/min-cut theorem [14] makes the cut value the optimum of a linear program. This section realizes that analogue exactly.

At nondegenerate seeded capacities, every tested boundary region has a unique minimum cut, with an exported margin to the next-best cut. The max-flow primal and the cut dual are exported as matrices, and exact certificates verify strong duality and complementary slackness. Writing y_e for the dual variable attached to the capacity c_e of edge e ,

$$\text{Area}(\text{mincut } A) = \text{OPT}_{\text{dual}}(A) = \sum_e c_e y_e.$$

The per-edge *shadow prices* are the y_e , not the optimum itself. Within a chamber of capacities where the minimum cut is unique, the y_e are the 0/1 indicators of the cut edges and equal the sensitivities $\partial F^*/\partial c_e$, which are verified by two-sided perturbations.

Entropies of contracted states. Random tensors placed on the same graphs and contracted independently give boundary states whose entropies obey $S(A) \leq \text{mincut}(A)$ on the sampled carrier. The mean ratios $S/\text{mincut} = 0.729, 0.904, 0.941$ over five seeds at bond dimensions $D = 2, 3, 4$ are a sampled finite trend toward saturation. They do not show convergence or equality, and

every sampled entropy stays below its cut. The min-cut entropy vectors satisfy monogamy of mutual information [15], and a fairly compared GHZ state [16] violates it. Together these support a **finite-graph recognition** of RT-type structure, conditional on the graph and tensor-network model. They do not derive holography, the coefficient $1/4G$, or a continuum law.

Two extensions that fail. Both failures are informative. First, there is no linearized-Einstein match. A pass can be manufactured at a degenerate base point, but at an honest nondegenerate point an independently specified geometric deformation does not reproduce the contracted-state δS under the declared pairing; uniqueness margins and a can-fail deformation are exported. Second, the Born and area ledgers do not obey one composition law. Area follows min-plus composition on all four tested gluing probes, while the Born ledger fails all four. What the two sampled ledgers share is only the monogamy inequality class, with different dependencies (review_2026/repairs/q5_lp_duality/).

5.5 Summary

question	status	scope
route mismatch	exact null result	the completions commute; 0.5 is an encoding-dependent table distance
fork, not ladder	exact toy theorem	coordinate-partition non-factorization on the declared Boolean carrier
field provenance	conditional finite construction	54 records with named coarse functionals and stated limitations
common carrier	hypothesis	semiclassical co-sourcing is a motivation, not a result
$T_{\text{QG-NoGo}}$	narrow	two special fused or stapled routes only
min-cut/LP identity	finite-graph recognition	exact dual certificates and sensitivities on non-degenerate finite graphs
contracted-state entropies	sampled	bound holds on the sample; saturation trend only
linearized Einstein	negative result	no match under the declared deformation pairing
one Born–area ledger	negative result	area passes four min-plus gluings; Born fails all four

6 Three construction programs

Three questions arise naturally from the framework as candidate predictions:

- **P1:** can identical entanglement (min-cut) data hide physically different geometries?
- **P2:** must gravity act as a classical record channel, so that it cannot entangle two masses?
- **P3:** does the stability of records force the absence of proton-decay and monopole mediators?

None of the three is forced by the machinery built here. Each is therefore developed as a *construction program*, which separates reproducible finite statements from what remains to be built. P1 has two finite subproblems, both completed below; its broader question remains open. Falsifiability language is used only for a specified finite model or a conditional channel, never for nature when the bridge to nature has not been built.

6.1 P1: entanglement and geometry

Where the question starts. The first carrier had fourteen edge capacities: eight terminal leaves and three two-edge bridges. Its 11,049-component min-cut fingerprint had a numerical Jacobian of rank nine, leaving five apparently invisible directions. Exact series-parallel reduction explains all five. The six bridge capacities enter only through

$$K = \sum_{i=1}^3 \min(w_{LM_i}, w_{M_iR}),$$

and the nine effective coordinates (eight leaves and K) have full rank nine on all three seeds. The apparent kernel was a redundancy of the parametrization, not evidence that different geometries share one fingerprint.

An exact search. The exact probe replaces finite differences by the exact active-cut incidence matrix: for each boundary region it enumerates the unique minimizing cut, records its margin, and computes the rank in rational arithmetic. Five exact local reduction classes are then applied: contraction of saturated terminals and zero columns; contraction of inseparable vertices with parallel sums; series reduction; replacement of an exact two-terminal module; and Δ -Y/Y- Δ replacement. The last move is accepted only if complete re-enumeration preserves every rational boundary cut value and the uniqueness of every minimizer. Among 378 weighted carriers, under a declared three-round Δ -Y search, 365 reduce to full rank. Thirteen remain, with residual-deficiency histogram $1^8, 2^4, 3^1$. Two candidate “sixth moves” (a transportation redistribution on a $K_{2,3}$ four-cycle and an opposite-perfect-matching redistribution on K_4) explain none of the thirteen survivors. This does not prove that the survivors are irreducible: exact graph quotients have not been exhausted, and the two candidate moves are only diagnostics (review_2026/probes/p1_kernel_quotient/). The census itself is falsifiable by re-enumeration: a different rank, cut value, uniqueness result, or reduction path would refute it. Two routes then push the question further, with artifacts and byte-comparing validators under open_programs/.

Route 1: graph level. The established statement is:

On 13 fixed exact-rational graph topologies, 19 kernel directions give nondegenerate intervals of positive capacities with identical complete terminal min-cut fingerprints and unique active minimizers. The selected endpoints have different weights and are not related by any weighted automorphism that fixes the terminal labels. For each of the 19 base/perturbed pairs, a queue-exhaustive forward search, using four directed reduction rules and Δ -Y/Y- Δ in both directions, finds the two sets of reachable presentations disjoint. The search admits only intermediate presentations with the same complete fingerprint and unique active minimizers, and it identifies presentations up to exact weighted isomorphism fixing the terminal labels.

This establishes disjointness of *forward* reachability only. It says nothing about orbits under the symmetric closure of the moves, about completeness of the move set, or about irreducibility. Indeed, an exact inverse-series witness shows that the symmetric closure is strictly larger than the computed forward sets. Series removal has a unique normal form on admissible subdivisions of the 13 pinned series-reduced carriers, but an exact counterexample, a subdivided leg of a Y interacting with a Δ -Y move, refutes the projection lemma that would have extended the result to the symmetric closure. That counterexample neither connects nor separates any of the 19 pairs.

Orbit disjointness under the symmetric closure therefore remains open, as do transformations that are inverse, context-dependent, nonlocal, or pass through degenerate intermediates.

Route 2: state level. The second route asks whether two tensor networks with the same graph and the same cut data can produce different boundary states. A finite engine computes the states exactly. Six exact-algebraic pairs were constructed on the same graph, each with identical complete cut values and identical boundary state: the joint map from capacities to (cut data, state) is not injective. The question is then whether such pairs are genuinely different presentations or merely gauge copies. Cut-preserving Δ –Y moves are not state gauge, since one can preserve every cut value while changing the state. But a diagonal intertwiner on the internal bonds can be constructed explicitly, and it meets the declared admission criterion for tensor-presentation gauge. The general statement is:

Gauge-collapse lemma. On any fixed, nonempty, connected binary-copy C2_L1 carrier with positive capacities, the normalized boundary state depends on the capacities only through the product of all edge capacities. Any two assignments with the same product are related by an exact diagonal internal-bond intertwiner, up to vertex-wise scalars.

The proof is an incidence-image argument. Because every vertex tensor copies its bit to all incident edges, a connected carrier supports only the all-zero and all-one internal assignments, so the normalized state depends only on the total product. For the intertwiner, the exponents of the internal-bond gauge must solve a linear system whose matrix is the oriented vertex–edge incidence map and whose target vector is indexed by vertices. On a connected graph, the image of that map consists exactly of the vertex vectors whose components sum to zero, and equal products make the target sum to zero. Connectedness is essential, since on a disconnected carrier a single global product does not control the separate components, and the certificate builder rejects such inputs. Carriers with zero or one edge are covered trivially (with one edge, equal products force equal capacities). The six constructed pairs are exact instances of the lemma, and exact regressions cover every connected labelled simple graph on up to four vertices (44 graphs) with orientation controls. The pairs are therefore exact non-injective fibers of the joint map but *not* inequivalent tensor-network presentations. Their separation by an exact invariant $I(c)$ survives only under a stricter relation that also preserves capacity labels. No physically complete bulk gauge relation and no theorem about underdetermination of bulk geometry is established.

6.2 P2: does gravity have to be a classical record channel?

Proposed tabletop experiments test whether gravity can entangle two masses [17, 18]. The finite calculation here is correct under an explicit premise. Suppose the gravitational mediator is a perfect classical record channel: it measures a geometry-visible label, stores it in orthogonal records, and conditions local responses only on that label. Then the induced two-mass channel is LOCC. Exact density-matrix calculation gives zero negativity and no CHSH violation, while a coherent-control channel does entangle. This is a standard conditional consequence of the assumed channel.

What is missing is the step that would force the premise. Neither the access structure of Section 5.2 nor the narrow fused-route statement derives a record channel. A geometry-diagonal coherent countermodel preserves the declared visible information while retaining phase coherence, and no mean-field guard stronger than a token-level check has been built. No prohibition on gravitationally mediated entanglement, and no forced null result for such experiments, is therefore claimed.

Within the assumed model, nonzero entanglement would falsify the implementation or the channel assumption. A positive experimental observation would rule out that channel as a model of the experiment, but it would not falsify a derivation, because this paper contains none. The open task is dynamical: derive, or fail to derive, record formation and decoherence from an explicit gravitational interaction without stipulating orthogonal records.

6.3 P3: record stability

The question is whether requiring stable records forces a clean branch, and with it the absence of proton-decay and monopole mediators. The record predicate $RS(C, \phi)$ on a branch is a conjunction of persistence, distinguishability, and a capacity condition: at least θ record tokens of the declared kind. The reference threshold is $\theta = 2$, and $\theta = 1, \dots, 4$ are reported as sensitivity checks. The comparisons below are made on the 12 admissible singleton branches, which lie over 8 of the 52 structures; the two-scalar branches are not part of this census. The answer depends on what counts as a record. Under a restricted record-token grammar, built from a chosen set of meson and epsilon generators, the implication holds nonvacuously for $\theta = 2, 3, 4$ (four clean branches satisfy RS, no breaking branch does) and fails at $\theta = 1$. Branch-complete ablations then show that the implication changes when the token census changes.

The most inclusive probe enumerates candidate invariant operators within a declared bound: fermion arity 2 to 4 (enough to cover the largest residual epsilon arity plus one), at most two scalar insertions ϕ or $\phi^\dagger$, and at most six constituents in total. A candidate is admitted only if two separately computed gates pass: exact singlet multiplicity under every UV factor and every unbroken residual factor, and zero charge under both the UV $U(1)$ and the residual $U(1)$ derived from the scalar’s stabilizer. The census must, for example, count the regression candidate $\epsilon(o_{2c0}, o_{3c0})$, whose candidate UV channel is $\epsilon o_2 o_3 \phi^\dagger$ with charges $-2 + 4 - 2 = 0$, and must exclude a spectator-Cartan token.

Under this complete bounded census, at $\theta = 2$, all three readings of the implication fail: pointwise ($RS(C, \phi) \Rightarrow \text{Clean}(C, \phi)$), universal over a structure’s branches, and existential over them. They fail both with and without scalar-dressed candidates; the existential failures are the four $SU(4)$ -only structures, which have selected breaking branches and no clean branch. The same failures occur at every tested threshold $\theta = 1, \dots, 4$. Among the tested definitions, the implication holds only under the restricted generator grammar, and there only for $\theta \geq 2$. The result is therefore **token-definition-sensitive**, and the completeness of the undressed token definition matters as much as the treatment of dressing.

These are counts of candidate channels, not of physical operators. For each candidate, the channel count is the smaller of the separately computed UV and residual singlet dimensions; this is only an upper bound on the rank of a UV-to-residual restriction map. No such restriction map, nonzero image under the vacuum expectation value, Grassmann or equation-of-motion quotient, or independent invariant-operator basis has been constructed. The census contains 556 candidate lifts, which give 404 candidate labels once lifts with the same residual component multiset and channel index, differing only by insertions of the same branch VEV, are identified by declaration. That identification is a counting convention, not a certified equality of operators or of VEV images (review_2026/repairs/s6_record_grammar_ablation/).

Consequently, no result here excludes proton decay or monopoles, and observing either would not falsify a theorem established here. The open program is to define records dynamically and to test which invariant candidates actually persist under a branch’s evolution. Only such a persistence criterion could turn the algebraic census into a physical statement.

6.4 Status summary

program	status	established statement
P1, graphs	finite subproblem completed	19 exact equal-cut pairs have disjoint forward-reachable sets; symmetric-closure orbit disjointness open
P1, states	finite subproblem completed	six exact joint-map fibers; all gauge-equivalent by the general C2_L1 lemma; no bulk-geometry theorem
P2, BMV	conditional; forcing open	an assumed perfect-record channel is LOCC and entanglement-null; deriving the channel is open
P3, records	open	on the 12 singleton branches at $\theta = 2$, the bounded candidate census is token-definition-sensitive; dynamical persistence is open

7 Further applications

This section applies the same vocabulary to four further puzzles. The applications differ in strength and are graded individually: the Bell row recovers a standard result, the information-loss row rests on a general theorem with exact certificates, the cosmological-constant row is sample-bounded numerics, and the measurement row is a conditional lemma with a stipulated two-example taxonomy. They show that the vocabulary travels; they are not evidence of universality.

7.1 Bell nonlocality: the common carrier is not a hidden variable

The common-carrier hypothesis of Section 5.3 invites an obvious objection: a single carrier underlying both readouts sounds like a local hidden variable, and local hidden variables are ruled out by experiment. The calculus turns the objection into a computation.

For a Bell pair with the standard CHSH settings [19, 20] ($a_0 = Z$, $a_1 = X$, $b_{0,1} = (Z \pm X)/\sqrt{2}$), the correlations computed from the state give $\text{CHSH}_{\text{quantum}} = 2\sqrt{2} \approx 2.8284$. Enumerating all 16 deterministic local strategies gives a maximum of exactly 2, and since CHSH is linear in the correlations, no mixture of those strategies does better. This is Bell’s theorem in its usual CHSH form, recovered, not discovered. A classical mixture with $\text{CHSH} = \sqrt{2}$ serves as the control: an explicit 16-row local table reproduces its correlations to 1.8×10^{-15} .

The repository files this result in the form of the F49 law. F49 asks whether a correlation table factors deterministically through a common source fiber, through an interface, or through a declared direct route between the wings, each with its own admissibility conditions. Formally, a *nonlocal residual* requires that all of these explanations fail *and* that a declared locality package fail. The finite construction records the failure of the declared common-source and interface explanations for the Bell table; it does not construct the full set of deterministic maps, admissible classes, and locality-package failure that a formal F49 nonlocal-residual instance would need. The F49 label is therefore bookkeeping for the standard result. The probabilistic content comes from the CHSH bound, not from F49, which concerns deterministic factorization and is not a probabilistic Bell theorem.

The calculus-specific point concerns the carrier. A local hidden variable would have to be an *admissible* common source for the complementary settings. F37 says that complementary accesses have no joint quotient within a declared admissible class. (The paired map (q_A, q_B) always exists as a mathematical map; what fails is its admissibility as a joint access.) The computed normalized

commutator residuals of the settings, $1/\sqrt{2} \approx 0.707$ under the declared normalization, match the carrier's complementary-pair control. Hence the following **forbidden rule**: a common mathematical carrier sitting above the access structure is not a Bell-local hidden-variable mechanism, and any reading of the framework that uses it as one is excluded.

7.2 Information loss: a property of the readout

The F34 law asks whether information is lost along a map. Given a transport $\tau : H_0 \rightarrow H_1$, a readout ρ on H_1 , and source information σ on H_0 , consider

$$\mathcal{O}_\rho = \{(h, h') : \rho(\tau h) = \rho(\tau h') \text{ and } \sigma(h) \neq \sigma(h')\}.$$

If $\mathcal{O}_\rho$ is empty, σ can be recovered from the readout (as a function on the image of $\rho \circ \tau$); if not, information is lost. We apply this to a toy holographic carrier, with no dynamics and no evaporation, asking where information loss lives. Figure 6 summarizes the answer.

The carrier. An interior configuration is a pair of real tensor frames (L, R) , each 81×27 , joined along an internal bond of dimension $d = 27$. The transport τ contracts the bond to the raw boundary matrix $M = LR^\top$. Internal gauge transformations $(L, R) \mapsto (LG, RG^{-\top})$, with $G \in GL(d)$, leave M unchanged. “Physical” interior information means gauge-invariant functions of the pair.

Theorem 7.1 (Full-readout fibers are gauge orbits). *Let K be a field, $L \in K^{m \times d}$, $R \in K^{n \times d}$, and $M = LR^\top$ with $\text{rank } M = d$. If $L' \in K^{m \times d}$ and $R' \in K^{n \times d}$ satisfy $L'R'^\top = M$, then there is a unique $G \in GL(d, K)$ with $L' = LG$ and $R' = RG^{-\top}$. Conversely, every such pair satisfies $L'R'^\top = M$.*

Proof. Since $\text{rank } M = d$ and every factor has d columns, L, R, L', R' all have full column rank. Because $R^\top$ is onto K^d , the column space of M equals that of L , and likewise that of L' . The columns of L and of L' are therefore two bases of one d -dimensional space, so $L' = LG$ for a unique invertible G . Multiplying $LR^\top = LGR'^\top$ on the left by any left inverse of L gives $R^\top = GR'^\top$, that is, $R' = RG^{-\top}$. The converse is $LG(RG^{-\top})^\top = LR^\top$. $\square$

Hence, at full rank, the fiber of the raw boundary readout is exactly one gauge orbit, and every gauge-invariant function of (L, R) factors through the readout on its image. Whether the theorem applies to the actual carrier is a question of exact rank, which floating-point singular values cannot settle. It is settled by certificate: for each of the three seeds, both frames, with their double-precision entries read as exact binary rationals, are reduced modulo the prime $2^{31} - 1$ and have a nonzero 27×27 minor. Each frame thus has exact rank 27, and so does the exact product M . (This does not certify the rank of the *rounded* floating-point product.) The rank hypothesis is needed: with $d = 2$, the pairs $L = \text{diag}(1, 0)$, $R = I$ and $L' = R' = \text{diag}(1, 0)$ contract to the same nonzero matrix, yet R and R' have different ranks, so no gauge transformation relates them.

One subtlety matters for physics. A quantum state is a normalized ray, not a raw matrix. If the readout is the normalized boundary state, its fibers are gauge orbits only after an overall scalar is also allowed, and only functions invariant under both the gauge and that rescaling can be recovered. For example, $\|M\|^2$ is gauge-invariant but cannot be recovered from the normalized ray, since $(2L, R)$ gives the same ray with four times the squared norm.

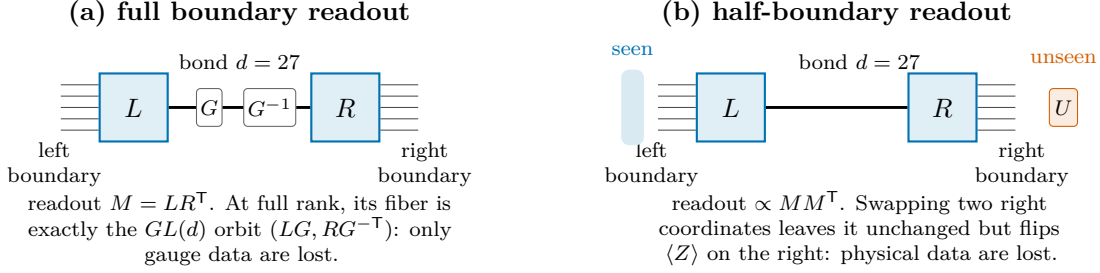


Figure 6. Where information loss lives in the toy holographic carrier. (a) Inserting GG^{-1} on the internal bond never changes the boundary matrix, and by Theorem 7.1 nothing else does at full rank: what the full raw readout forgets is pure gauge (for normalized states, gauge plus an overall scale). (b) A readout of the left half alone cannot see a permutation U of the right half, yet that permutation changes a right-boundary expectation value. Information loss is a property of the readout, not of the carrier.

Partial readout. Now let the readout be only the reduced state on the left half of the boundary, the toy analogue of seeing only the early radiation. Replace R by UR , where U is the permutation exchanging the first two right-boundary coordinates. Then $M' = MU^T$ and $M'M'^T = MM^T$, so the left reduced state is exactly unchanged. But the right-boundary observable $Z = \text{diag}(1, -1, 0, \dots, 0)$ has expectation -4.674×10^{-3} before the swap and $+4.674 \times 10^{-3}$ after it, computed exactly in integer Gram arithmetic for seed 101. The two full normalized states therefore differ while the partial readout agrees: physical information is genuinely lost.

Forbidden rule. In this toy carrier, loss of physical (gauge- and scale-invariant) information is excluded at the full boundary readout and forced at the half-boundary readout. The information puzzle [21] is, here, a property of the readout rather than of the carrier. No dynamics, evaporation, or Page curve [22] is modeled, and nothing follows about real black holes (review_2026/repairs/f34_exact_factorization/).

7.3 The cosmological constant: no in-layer value law

The F50 carrier contains an analogue of a cosmological background [23]: the matter-independent term V_0 in $V[\psi] = V_0 + \kappa T_{00}[\psi]$. A fixed solver budget (60 iterations at mixing 0.5) converges for only 15 of 18 sampled background offsets (it fails at offsets -5 , -3 , and 10), which looks like an admissibility boundary. It is an artifact of the budget: at 120 iterations all 18 converge. A historical finite-budget boundary in κ is likewise not an existence boundary, since all 12 sampled κ values have fixed points. Adaptive iteration with damping values $0.2, 0.35, 0.5, 0.7$ and at least three initial states finds a fixed point for every sampled background and every sampled κ , so there is no existence boundary on the declared sample. The Jacobian of the raw (undamped) fixed-point map offers a different, non-unique discriminator: its spectral radius crosses one near background offset 3.6535 and near $\kappa = 5.8675$, and 15 of the 18 sampled backgrounds are linearly stable under that map. Whether that criterion defines physical admissibility is a modeling choice. The sample therefore supports only this: *no unique value law for Λ is selected within the layer, and every sampled background admits a fixed point*. These are sampled numerics, not interval-certified existence or bifurcation theorems (review_2026/repairs/q7_f50_convergence/).

7.4 The measurement problem: a conditional discriminator

The established content is a definability lemma, conditional on a supplied source language and closure Σ_f on the declared carrier, together with a two-example taxonomy that is stipulated to

separate “ledger-changing” from “readout-changing” accounts of measurement. It is not a theorem that strictly extends quantum mechanics, and it is not an operational test for future measurement theories. A generated class of theories and a physical correspondence would be needed before that language is warranted.

7.5 Summary

puzzle	law	verdict	control
Bell nonlocality	F49, F37	CHSH $2\sqrt{2} > 2$ recovered; the carrier is not a local hidden variable	classical mixture with an explicit local table
information loss	F34	full raw readout loses only gauge data (theorem); half readout loses physical data	rank-deficient pair; exact permutation witness
cosmological constant measurement	F50, F47/F26 non-factorization	fixed points for every sampled background; no unique in-layer value law Σ_f -conditional lemma; stipulated two-example taxonomy	raw-map stability crossing, if adopted identical-diagonal witness pair

These rows are bounded diagnostics on their declared carriers. They add neither established predictions nor a universality result.

8 What the two tracks share

Both tracks use the same formal vocabulary: quotients, closure, factorization defects, common refinements, and graded results. That reuse is real, and it is useful for organizing finite constructions. It does not by itself support stronger statements: that the grammar *forces* physics across domains, that it reaches a foundational grade, or that it is universal. Stronger ties would need constructions that are not available here: a grounding of record stability, a derived gravitational record channel, a non-commuting pair of routes, or one composition law shared by Born and area ledgers.

8.1 Keeping the tracks separate, and the limits of that

The SM construction was typed as a selection layer and the QM–GR construction as a common-refinement problem. Validators rejected certain cross-track tokens, and the build history contains genuine self-corrections. This is evidence of procedural separation, not of mathematical independence or blinding. The protocol was designed and reported by the same program, an earlier external gate on the missing-layer atlas never issued final approval, and the atlases are not independent preregistration evidence. A formal object that recurs in both tracks is therefore a comparison datum, not proof that nature realizes a substrate-independent law.

8.2 An inventory of shared objects

formal object	Standard-Model track	QM-GR track
factorization through a readout	clean/breaking classification of branches on the finite carrier	exact incomparability of the declared coordinate readouts; no commutator interpretation
common refinement	the regular $SU(5)$ embedding constructs hypercharge and coset roles under named inputs	the abstract carrier L factors both readouts by construction; field provenance is conditional
record vocabulary	bounded censuses of candidate invariant channels; the implication is token-definition-sensitive	an assumed perfect-record channel is LOCC; record formation is not derived
value and moduli diagnostics	the F47 surface is convention-sensitive and selects no measured scale	every sampled F50 background has a fixed point; raw-map stability is an optional criterion, not a Λ law
finite graph duality	no cross-track identity	exact min-cut primal/dual certificates and shadow prices; entropy saturation only sampled

The table supports a modest statement: the same typed questions expose analogous failure modes on two deliberately different finite substrates. It does not identify the physical mechanisms. In particular:

- the $SU(5)$ X/Y coset has several computed roles inside one constructed frame, but it is not the QM-GR factorization defect and does not explain perturbative non-renormalizability;
- record candidates in the gauge carrier and a classical-record quantum channel share terminology, but no map connects them and neither forces the other;
- commuting diagrams of the F51 shape organize both an embedding and an access fork, but this formal resemblance does not select $SU(5)$, prove that the QM-GR parent is unique, or establish a universal composition law.

8.3 The one-grammar statement, bounded

What can be claimed across the tracks is this:

One frozen structural vocabulary was instantiated on two finite, differently typed carriers. It produced several sound conditional constructions, two genuine toy theorems, and useful diagnoses of where stronger claims fail. The recurrence is a methodological datum with $n = 2$. It neither demonstrates universality nor transfers any statement to nature.

The bound is what makes the negative findings informative rather than embarrassing. Exact quotienting shows that the naive P1 kernel was a reparametrization, and P1's two finite subproblems are then completed at a precisely stated strength: forward-reachability disjointness for nineteen graph pairs, and gauge collapse of the six state-level pairs. Explicit channel analysis isolates the missing dynamical premise of P2, and a broader invariant census exposes P3's dependence on the definition of a record. P2 and P3 remain construction programs that require gravitational dynamics and invariant-ring and record machinery not supplied here.

Read as a whole, the development recorded in Appendix C is a bounded example of theory repair guided by the calculus: typed failures led to reconstructed carriers, sharper theorem objects, exact countermodels, and precise successor requirements. It proposes no new physical theory of

the SM, QM, or GR. What it demonstrates is a workflow for engineering and auditing emergence claims, at the scale of one research package.

8.4 Limits

- Both carrier families are finite and deliberately capped; there is no continuum limit and no transfer to nature.
- The separation of the tracks is procedural self-report, not independent blinding.
- Recognition results keep every named input, in particular the regular $SU(5)$ embedding and the graph and tensor-network model.
- The results passed an internal adversarial claims audit with machine-verifiable validators; that is not journal review or independent replication.
- The missing-layer and unification atlases have unresolved limitations as public support (Section 9 and Appendix B).

9 Scope, limits, and testability

Every result in this paper is finite and conditional: either an exhaustive or sampled computation on a finite carrier, or a general theorem about a fixed finite graph or fixed-size matrices whose finite instances are checked by validators. A validator can falsify a reported table or an exact identity, but that does not make an extrapolation from a carrier to nature valid. Conversely, a physical observation cannot falsify a forcing theorem that the paper does not contain. Table 6 collects each established statement with its grade and its most direct test.

Table 6. Established statements, their grades, and their most direct tests.

statement	grade	bound, or most direct falsifier
selection layer (L^*)	coherent finite construction	one-layer hypothesis and within-layer block structure; a constructed admissible selector outside the layer would test it
gauge census	finite enumeration	exact alphabets, caps, and equivalences; independent regeneration can falsify the counts 1,066/419/195 and survivors 52/24/24
branch selection	conditional finite selection	singleton and two-scalar caps; “every branch clean” fails, while the clean branch class is unique to $2 3$ and $SU(2)$ -active
single-factor exclusion	toy theorem	stated grammar only; the validator rebuilds artifacts and constructs witnesses for $N = 3, \dots, 12$
$SU(5)$ ratios and coset	exact conditional construction	regular embedding, $\bar{\mathbf{5}} + \mathbf{10}$, weak-pair normalization; not a parent selector or low-energy prediction; Pati–Salam also gives $3/8$
content orbit and mass ranks	recognition within template	declared quotient and renormalizable one-scalar grammar; $N = 1, \dots, 4$ only
coordinate incomparability	exact toy theorem	16 coordinate-subset partitions, 55/120 incomparable pairs; field provenance is a separate conditional construction

statement	grade	bound, or most direct falsifier
min-cut/LP identity	finite-graph recognition	unique-cut chambers, exported primal and dual matrices, exact slackness and sensitivity checks
contracted-state entropies	finite sample	bound, $D = 2, 3, 4$ trend, and monogamy with GHZ control; no convergence or equality
information loss (F34)	general theorem with exact certificates	full-rank stratum and raw readout (normalized rays add an overall scale); exact rank certificates for three seeds; exact half-boundary witness; no dynamics
F50 backgrounds	sample-bounded numerics	every sampled background and κ has a fixed point; the stability boundary depends on the criterion
P1, graph level	finite forward-reachability result	19 exact equal-cut pairs on 13 carriers; forward-reachable sets disjoint; symmetric-closure orbit disjointness open
P1, state level	exact noninjectivity and gauge collapse	six connected C2_L1 joint-map fibers; all collapse under the general diagonal-gauge lemma; no bulk-geometry theorem
P2, BMV	conditional LOCC statement	null only for an assumed perfect-record channel; record formation not built
P3, record stability	open program	bounded candidate census is token-definition-sensitive; restriction maps and dynamics not built
singularity row (G2)	finite toy	an existential regularized example; finiteness comes from a saturating rule, not from physical singularity resolution

9.1 Supporting atlases

The repository also contains three card atlases, which are not evidence for the claims above. The missing-layer atlas is **unfit** as independent preregistration or blinding evidence: its card grades exceed the established claims, and the external gate on it never gave final approval. The retrodiction atlas is **fit with findings**: its recovery and reframing grades are honest, and it claims no new retrodiction. The unification atlas is **scientifically fit with findings but mechanically incomplete**: its ledger of tensions is useful, but 19 of its 34 candidates have only explicitly typed missing-card coverage and need completed cards before they can serve as public support. Appendix B lists the remaining validator caveats.

9.2 What can be tested now

The most direct tests are computational: regenerate the gauge carrier and branch table; reconstruct the exact $SU(5)$ commutant and coset action; materialize the N -copy mass matrices; verify the coordinate-partition lattice; re-solve the min-cut primal and dual; recheck the F34 rank certificates and the half-boundary witness; repeat the adaptive F50 grid; and rerun the P1 forward-reachability and gauge-collapse validators. The open parts of the programs (symmetric-closure separation and the broader physical reading of P1, and the P2 and P3 constructions) may eventually be answered

in ways that disagree with the motivations given here, without contradicting any established finite statement.

10 Discussion and outlook

10.1 Methodological lessons

The strongest lesson of this program is methodological. A finite computation can be exact and still answer the wrong question. Aliases can corrupt a denominator. A witness for one branch can be mistaken for a property of a structure. A distance between tables can be mistaken for a commutator. A null direction can turn out to be a reparametrization. A solver budget can masquerade as an existence boundary. A theorem about raw matrices can be read as one about normalized states. Each of these happened in the course of this work, and each result reported here is stated only after the relevant distinction has been made explicit.

The positive inventory is nonetheless useful. The gauge carrier yields a branch-level selection under declared caps. The regular $SU(5)$ frame and its coset are explicit exact objects. The content quotient produces one selected orbit, and the N -copy probe computes genuine generic ranks. The Boolean access lattice supplies an exact incomparability theorem; the min-cut carrier supplies an exact LP certificate together with independent contracted-state evidence; and the toy holographic carrier supplies a general statement of where information loss lives. None of these is a theory of nature, but each is a reproducible construction with a clear failure condition.

10.2 Nearest next steps

For P1, the next target is the symmetric closure of the graph moves. The obstruction to the obvious projection argument is now located precisely: the interaction of a subdivided Y-leg with a Δ -Y move. Handling that interaction is a necessary obligation for any such upgrade, not a sufficient one: an invariant that settles orbit disjointness must be preserved by every move in the claimed symmetric closure and must still separate the endpoints. Other tensor families and capacity conventions, and the broader physical or bulk-geometric reading, remain open.

Two further upgrades dominate. First, relevance to BMV-type experiments requires a dynamical model that derives record formation, rather than stipulating a measure-and-record channel. Second, the record-stability program requires the physical invariant algebra and its dynamics: UV-to-residual restriction maps and their ranks, nonzero images under the vacuum expectation value, Grassmann, equation-of-motion and syzygy relations, and a persistence criterion for candidate records.

Smaller upgrades are also worthwhile: rebuilding the historical multi-factor census counts on the final carrier, replacing the remaining summary-reading validators, establishing or refuting large-bond saturation of the min-cut bound with a rate, widening the representation and scalar windows, and generating a physically motivated class of measurement models in place of a two-example taxonomy. A third substrate could test whether the vocabulary stays useful, though it would not by itself prove universality.

10.3 What kind of framework is this?

As used here, the calculus is the finite diagnostic subset described in the introduction: obstructions, quotients, construction audit, and claim bookkeeping. The strict-extension, reclosure, promotion, endogenous-repair, and run-level dynamical parts of the Six Birds corpus are not used, and no

continuum bridge to a physical substrate is supplied. This subset makes hidden quotients and quantifiers visible. It cannot turn an unbuilt physical bridge into a prediction.

Several of the exact objects sit naturally at this level. The gauge-collapse lemma says that, on any fixed connected binary-copy carrier with positive capacities, equal total edge products imply an exact diagonal intertwiner on the internal bonds, up to vertex-wise scalars. The full-readout factorization theorem says that, at full rank, a boundary matrix determines its interior factors up to exactly the internal gauge. Both are general statements whose applicability to a given carrier is checked by exact certificate rather than by sampling.

10.4 An invitation

The most useful independent attacks are concrete: regenerate the gauge carrier under a different declared quotient; test the P1 pairs under transformations outside the declared move classes, or under other tensor families and capacity conventions; derive a gravitational record channel from dynamics; or compute the restriction maps and relations of the record invariant ring. A disagreement with any exported finite table is a direct reproducibility failure. A physical result bears on an open program only once the missing map from carrier to physics is supplied.

11 Conclusion

We asked whether one small vocabulary of quotients, obstructions, and common refinements can usefully organize two very different problems in fundamental physics. On declared finite carriers, it can, and it delivers specific objects. These are a conjugation-consistent, branch-level selection of $SU(2) \times SU(3)$; a witness-independent toy theorem excluding single-factor candidates; an explicitly constructed $SU(5)$ frame and coset that recover the standard unification ratios; a unique content orbit under a named quotient; generic mass ranks for $N = 1, \dots, 4$; an exact incomparability theorem for the QM and GR readouts, with conditional field provenance; a finite-graph min-cut/LP duality supported by contracted-state and monogamy controls; and a general theorem locating information loss in a toy holographic carrier in the readout rather than the carrier. Every statement carries its caps, conventions, named inputs, sample bounds, or depth limits.

The negative results are results in their own right, and they mark what the framework cannot presently claim. The quantize and curve completions commute, so this level offers no route-mismatch explanation of why quantum gravity is hard. The bounded record-stability census is sensitive to the definition of a record. Born and area ledgers do not share the tested composition law. The linearized-Einstein response does not match at an honest nondegenerate point. And the generation grammar neither stays blind to N nor selects $N = 3$.

Of the three construction programs, P1 has two completed finite subproblems. Nineteen exact equal-cut graph pairs have disjoint forward-reachable sets, with disjointness under the symmetric closure still open. Six exact state-level pairs with identical cut and state data turn out, by a general lemma, to be gauge-equivalent tensor networks. No bulk gauge relation or theorem about underdetermined bulk geometry follows. P2 and P3, dynamical gravitational record formation and dynamical persistence of records, remain open.

The common grammar has not forced the answers to these questions. Its demonstrated contribution is narrower and, we think, durable: it organized two finite construction programs and, under sustained adversarial scrutiny, showed exactly where stronger claims would fail.

A The calculus, formally (working subset)

This appendix states every formal object the body relies on, at usable precision. The full development is in Foundations II–IV [4, 5, 6].

A.1 Descent and obstruction

For a surjection $q : H \rightarrow Q$ and a readout $t : H \rightarrow T$,

$$t \text{ descends} \iff \mathcal{O}_q(t) = \{(h, h') : q(h) = q(h') \wedge t(h) \neq t(h')\} = \emptyset \iff \exists \bar{t} : Q \rightarrow T, \bar{t} \circ q = t.$$

On a finite carrier both directions are finite checks on $H \times H$: choose a representative in each fiber, and note that $\bar{t}$ is well defined exactly when t is constant on fibers. If q is not surjective, the same statement holds with $\bar{t}$ defined on the image of q .

A.2 The factorization defect (Foundations III, Theorems 12–13)

For maps $\pi_0 : S \rightarrow O_0$ and $\pi_1 : S \rightarrow O_1$ on a finite carrier,

$$\Delta_{\text{fact}}(\pi_0, \pi_1) = \{(s, s') : \pi_0(s) = \pi_0(s') \wedge \pi_1(s) \neq \pi_1(s')\}, \quad \Delta_{\text{fact}} \neq \emptyset \iff \nexists \phi : \pi_1 = \phi \circ \pi_0 \text{ on } S.$$

It is used for the branch-level clean-separation classification (Section 4.3), the single-factor toy theorem, and the QM–GR coordinate-partition defect with eight unordered witness pairs (sixteen ordered pairs) in each direction (Section 5.2). Generator action separately establishes the X/Y roles inside the constructed $SU(5)$ frame. These uses do not identify one physical object.

A.3 Non-commuting completions (Foundations III, Theorems 9–10)

There are idempotents E_1, E_2 on a finite carrier with $E_1 E_2 \neq E_2 E_1$, and the route-mismatch defect $\Delta_3^{\text{comp}}(E_1, E_2) = \{c : E_1 E_2(c) \neq E_2 E_1(c)\}$ is nonempty exactly when they fail to commute. A minimal witness: on $C = \mathcal{P}(\{a, b, c\})$, let $E_1(S) = S \cup \{b\}$ if $a \in S$ and $E_1(S) = S$ otherwise, and $E_2(S) = S \cup \{c\}$ if $b \in S$ and $E_2(S) = S$ otherwise. Then $E_1 E_2(\{a\}) = \{a, b\} \neq \{a, b, c\} = E_2 E_1(\{a\})$. This is a mathematical example only. The two conditional-mean completions constructed for QM–GR commute exactly (Section 5.1), so no link to non-renormalizability is established.

A.4 The six roles and the no-algebra theorem

$\mathbb{P} = \{P_1, \dots, P_6\}$ (descent, representability, route mismatch, refinement, packaging, audit) are labels for typed judgments. Foundations III proves that no total binary operation $* : \mathbb{P}^2 \rightarrow \mathbb{P}$ decodes the status of a judgment. The proof exhibits two cells with the same pair of primitives and different statuses; if status were $\text{decoder}(\text{op}(\text{pair}))$, the two statuses would coincide. The theorem does not say that no binary operation on six symbols exists: $\text{op}(x, y) = x$ is one. The corpus also makes no claim that all structure decomposes into the six. Directed-cell notation $P_i \leftarrow P_j$ names a typed, individually audited judgment, never a product. The relevant Lean declaration and its classifier countermodel build under Lean 4.28.0; that build does not establish any physical premise.

A.5 The F-laws used in this paper

Each law is a statement about quotients and closures that holds whenever its premises do. The third column names the premises that must be supplied or constructed before the law applies; the paper never treats those premises as consequences of the law.

law	statement (core)	premises to supply; use here
F23 probability	a readout unresolved on the fibers of q receives the stable weight $\Pr_{q,r}(a, v) = m(\lambda(a))(\{h : q(h) = a, r(h) = v\})$	a fiber-supported measure m and stability data (imported, not derived; the Born rule is not derived); Born ledger, Section 5.4
F24 role split	role split and refinement descend through the declared quotient	architecture-family names are declared statuses, not an exhaustive physical classification; Section 4.1
F26/F47 contingency, fine-tuning	no value law lands on a contingent (moduli) selection; fine-tuning means a small selector region	a declared measure and threshold; Sections 4.6 and 7.3
F27 conservation	a charge is conserved iff it descends through the orbit quotient	an actual supplied orbit relation; Section 4.4
F34 information loss	σ is recoverable from $\rho \circ \tau$ iff $\mathcal{O}_\rho = \{(h, h') : \rho(\tau h) = \rho(\tau h') \wedge \sigma(h) \neq \sigma(h')\} = \emptyset$, with recovery defined on the image of $\rho \circ \tau$ (or on a surjective declared codomain)	the transport, readout, and source information; Section 7.2
F37 complementarity	$q_A \perp q_B$ iff there is no joint quotient (J, j, a, b) in the declared admissible class with $a \circ j = q_A$ and $b \circ j = q_B$	the admissible joint class; the pairing (q_A, q_B) itself always exists as a map; Section 7.1
F39 entropy	$E(q)$ is a fiber volume, with chain rule $E(q_C) = E(q_F) \odot m$, monotone under coarsening	the mixing chain rule, monotonicity of the combination, and the zero/injectivity identification; it does not identify arbitrary physical entropy with fiber volume; area ledger, Section 5.4
F48 gluing	a global obstruction is a nontrivial gluing class	an actual gluing class; Section 4.4
F49 common source	a correlation table κ is locally explainable iff it factors deterministically through a common source fiber, an interface, or a declared direct route (each with its admissibility conditions); a nonlocal residual requires that all of these fail and that the declared locality package fail	the maps, admissible classes, and locality package; deterministic fiber factorization, not a probabilistic Bell theorem; Section 7.1
F50 background selection	a descended background b is <i>necessary</i> (b constant) or <i>moduli</i> ($b(m) \neq b(m')$ for some m, m'); it is a <i>vacuum</i> iff the admissible set is a single point $\{m_*\}$	the admissibility criterion; Section 7.3
F51 unification	a parent Q_U with commuting squares $\pi_A \circ q_U = q_A \circ \iota_A$ and $\pi_B \circ q_U = q_B \circ \iota_B$, plus status-compatible claim lifts	the parent and the maps; Sections 4.4 and 5.2

A.6 Grades and audit checks

Grades. COMPUTE: a value or verdict obtained from the construction. SELECT: a choice made on declared conditions. PROVE-BLIND: an invariance theorem showing that a quantity is an observed input. GROUND: a condition supplied as the shadow of a named higher source, with its warrant and strength caveats. RECOGNITION: a known result reproduced as an instance of a law,

non-circularly. FORBIDDEN RULE / DEMARCATION: a computed exclusion or classification with a can-fail control.

Audit checks. These are intended to be mechanical where implemented; validator coverage varies, with the exceptions listed in Appendix B. *Anti-tautology*: the target is not definitionally equal to its evidence. *Anti-circularity*: each hypothesis is satisfiable by an exhibited structure that violates the conclusion. *Frozen machinery*: imports are pinned by SHA-256 and not retuned. *Can-fail control*: a configuration on which the claim fails. *No overclaim*: artifact text is checked against a blocklist. *Anti-contamination*: tokens foreign to a track fail its build.

A.7 Key quantities

- **Embedding ratios** (Section 4.4). The exact regular- $SU(5)$ construction derives the one-dimensional hypercharge direction and decomposes $\bar{\mathbf{5}} + \mathbf{10}$. Under the weak-pair normalization, $\text{Tr } T_3^2 = 2$ and $\text{Tr } Q^2 = 16/3$, so $\sin^2 \theta_W = 3/8$; also $k_Y = 5/3$. These are not low-energy predictions, and Pati–Salam also gives $3/8$.
- **Min-cut/LP duality** (Section 5.4). For capacities c_e , exact strong duality gives $\text{Area}(\text{min cut}) = \text{OPT}_{\text{dual}} = \sum_e c_e y_e$. The y_e , not the optimum, are the per-edge shadow prices $\partial F^* / \partial c_e$ in a unique-cut chamber.
- **Monogamy** (Section 5.4). $I_3 \leq 0$ for min-cut-realizable entropy vectors, while GHZ_4 has $I_3 = +\log 2$.
- **CHSH** (Section 7.1). The local bound $\max_{\lambda \in \{\pm 1\}^4} |E_{00} + E_{01} + E_{10} - E_{11}| = 2$ is found by enumerating the 16 strategies; the Bell state with $a_0 = Z$, $a_1 = X$, $b_{0,1} = (Z \pm X)/\sqrt{2}$ gives $2\sqrt{2}$.
- **Full-readout factorization** (Section 7.2). If $M = LR^T$ has rank d equal to the bond dimension, every same-size factorization of M is (LG, RG^{-T}) for a unique $G \in GL(d)$. For normalized rays an overall scalar must also be allowed.
- **Active-cut quotient search** (Section 6.1). Exact 0/1 active-cut incidence matrices replace finite-difference Jacobians. The three-round probe leaves 13 of 378 weighted carriers unclassified; on those carriers, 19 exact pairs have disjoint forward-reachable sets under four directed reductions and bidirectional Δ –Y/Y– Δ , up to exact weighted isomorphism fixing the terminal labels. Symmetric-closure orbit disjointness and other transformations remain open.

B Constructions and reproducibility

The historical construction tracks live under `physics_atlas/`; the controlling repairs and probes live under `review_2026/`; the construction programs live under `open_programs/`. Where a historical headline and a controlling repair disagree, the repair and the claims registry `review_2026/CLAIMS_REGISTRY.md` control. Validators vary in strength: many historical scripts check deterministic artifacts, while the strongest rebuild their entire result in memory and compare every output byte for byte.

B.1 Entry points for the controlling results

result	artifact directory	validator (<code>-self</code>)
gauge carrier, branch census, scalar pairs	<code>review_2026/repairs/s1_carrier_reconstruction/</code>	<code>run_s1_carrier_reconstruction_v3.py</code>

result	artifact directory	validator (<code>-self</code>)
$SU(5)$ generators, ratios, coset, F27/F48	review_2026/repairs/s3_generator_construction/	run_s3_generator_construction_v2.py
content quotient, mass rank, N -copy probe	review_2026/repairs/s4_content_quotient/	run_s4.py
F24/F47 architecture and surface	review_2026/repairs/s5_f24_f47/	run_s5.py
record-token census and ablations	review_2026/repairs/s6_record_grammar_ablation/	run_s6_record_grammar_ablation_v3.py
field provenance and partition lattice	review_2026/repairs/q1_access_provenance/	run_q1.py
route-mismatch commutation	review_2026/repairs/q2_route_mismatch/	run_q2.py
min-cut LP duality, response, composition	review_2026/repairs/q5_lp_duality/	run_q5.py
adaptive F50 convergence and stability	review_2026/repairs/q7_f50_convergence/	run_q7.py
F34 factorization theorem, rank certificates, partial-readout witness	review_2026/repairs/f34_exact_factorization/	run_exact_factorization.py
five-class P1 quotient search	review_2026/probes/p1_kernel_quotient/	run_active_cut_quotient_v3.py
P1 graph level: forward reachability	open_programs/prog3_cut_fingerprints/step2_orbit_saturation/	run_step2.py
P1 graph level: symmetric-closure obstruction	open_programs/prog3_cut_fingerprints/step3_symmetric_closure/	run_step3.py
P1 state level: finite C2_L1 classification	open_programs/prog2_state_underdetermination/step5_family_gauge_classification/	run_step5.py
P1 state level: gauge-collapse lemma	open_programs/prog2_state_underdetermination/step6_exact_gauge_collapse/	run_step6.py

Each directory contains its design choices, literal dependency pins, schema, result tables, and a findings note; the caps and conventions stated in the body are also machine-readable there. The F34 directory additionally contains the written proof of Theorem 7.1 (PROOF.txt). The proof is general linear algebra; the Python code checks exact finite instances and is not a Lean mechanization of the universal statement.

B.2 Independent checks

The mathematics review in review_2026/mathematics_audit_20261003/ adds checks that do not reuse production code: the weight-character representation oracle of Section 3.1, tampering controls for the strengthened validators, regressions of the gauge-collapse certificate on every connected labelled simple graph with at most four vertices, and a per-claim coverage file for all 42 registry records. They are reproduced by

```
python3 review_2026/mathematics_audit_20261003/run_checks.py \
--output /tmp/sm-qm-gr-audit.json
```

Adding `-all` also sweeps the controlling and historical validators. The runner works in separate scratch copies built from tracked, non-ignored files, so a missing local artifact cannot silently supply evidence.

B.3 Historical track rebuilds and an external corpus variable

Step validators in a clean checkout resolve repository-internal sources relative to their own files. The usual pattern is

```
cd physics_atlas/thread_<track>/steps/<step_dir>
python3 run_stepNN.py --self
```

Four historical QM-GR build drivers (steps 51–54) also reproduce hashes of Foundations III/IV corpus .tex files that are not distributed here. Rebuilding those rows requires an explicit external root,

```
SIX_BIRDS_PAPERS_ROOT=/path/to/six-birds-papers \
python3 <step51--54-build-driver>.py
```

Without it, those drivers stop before writing partial artifacts and name the missing file. The variable is not needed to validate the released artifacts or to run repairs that only import compute functions; it is a provenance dependency, not evidence that this checkout contains the external corpus.

B.4 Validator caveats

The repository is not uniformly self-verifying, and it is not a tamper-evident freeze.

- **Summary-reading validators.** The validators of the historical multi-factor window-closure steps (steps 43 and 44 of the SM cluster) and some analogues check stored audit tables instead of rebuilding every load-bearing computation. The step-43 numerical census counts are also stale relative to the final carrier: a current-engine probe supports its $N \leq 6$ family conclusion, but a final-carrier rebuild remains to be done. By contrast, the step-61 validator (single-factor theorem) rebuilds every released non-code artifact in a temporary directory, compares bytes, and constructs the actual witness pairs.
- **In-place regeneration.** Some `-self` validators regenerate artifacts in place, so they are not idempotent checks of a release tree. A race in a parallel pool was observed once. Run sweeps in isolated scratch copies.
- **Pins.** The historical step-69 score source is pinned by a literal reviewed digest, so changing the source fails even if the stored conclusion is left unchanged. That step’s record implication is superseded by the record census of Section 6.3. The preregistration SHA256SUMS primer entry names a file absent from the release, and the freeze gate is a consistency check, not a cryptographic or externally timestamped freeze.
- **Tracked inputs only.** The QM-GR step-27 and step-28 validators reconstruct their operators and histories from tracked inputs (step 28 checks every stored history state and potential), and the evidence log required by the P1 state-level step-6 validator is tracked. The compatible-clock constraints of steps 27–28 are finite constructions whose compatibility is engineered, not a derived clock dynamics.
- **P1 search scope.** The initial P1 probe used a declared three-round Δ -Y/Y- Δ search. The controlling validator rebuilds nineteen exact fibers and exhausts their forward reachability under four directed reductions and bidirectional Δ -Y/Y- Δ , up to terminal-label-fixed exact weighted isomorphism. Symmetric-closure orbit disjointness, paths through tied-minimizer presentations, and other transformations remain open.

The Q5 replay policy separates exact discrete artifacts, which are compared byte for byte, from floating-point summaries, whose numeric fields use tight declared tolerances while all non-numeric

structure stays exact (review_2026/repairs/q5_lp_duality/reproducibility_policy.md). The registry records, for every claim, its wording, status, evidence type, supersession history, controlling source, and validator.

B.5 Paper build

The modular manuscript builds with `make paper-build`, which compiles `paper/main.tex` and writes `paper/build/main_flat.tex`; the tracked root `.tex` file is copied from that flattened output. All figures are drawn in TikZ inside the source, so the flattened file needs no external image files. The Qeios bundle is regenerated from the same sources.

C Audit trail: what was rejected, and why

We publish the program’s rejected constructions on purpose. A framework as expressive as this one can manufacture results in either direction, toward novelty or toward the safety of mere recovery, and the only durable defense is a record showing both failure modes being caught. Every entry below was detected by the checks of Section 2.4 and Section 3, or by line-by-line review against them, and rejected. Some replacements were themselves overturned later; this appendix keeps that chronology instead of presenting every intermediate replacement as final.

C.1 Tautologies: the target equal to its evidence

rejected construction	defect	how it was caught	replacement
area–entropy first form	“law”, area defined as a boundary cell count and S as $\text{cells} \times \ln 2$: equal by construction	reading the build: no can-fail case was possi- ble	the saturable bound $S(A) \leq \text{mincut}$ with strict-gap controls
area–entropy second form	“law”, bulk dimensions set equal to state dimensions, forcing the match	same	same
Born–area ledger”	“one the F39 entropy from the Schmidt spectrum and the F23 entropy from the eigen- values of ρ_A : the same spec- trum twice	agreement to 8.9×10^{-16} , an identity true for ev- ery pure state	the geometric min-cut as the F39 side: fixed under reseeding ($\log 3$) while S varies; the validator now fails on machine-precision equality
first-law quance”	“conse- $\delta S = \delta \langle H_{\text{mod}} \rangle$ restated on a density matrix disconnected from the carrier	correct but empty: noth- ing could have failed	the entropy-cone and monogamy consequence with the GHZ violator

C.2 Circularity and hardcoding: a load-bearing clause doing no work

rejected construction	defect	how it was caught	replacement
monogamy “forcing” (F51)	“forced = compatible and $I_3 \leq 0$ ” with the compatibility flag constantly true, and the broken-compatibility control written as literal booleans	the forcing reduced to the known fact that holographic states obey monogamy; the control was never computed	a computed compatibility control; a later composition probe refutes a universal shared Born-area law
clean-separation “grounding” (SM)	the internal grounding assumed clean separation	anti-circularity: a hypothesis not satisfiable by a violating structure	a record-stability construction, later shown token-definition-sensitive by the complete candidate census
SM “candidate-law generation”	a <code>total_slots = 5</code> prior and an exterior, GUT-shaped predicate smuggled the answer	adversarial review and internal re-audit	the conjugation-consistent carrier and branch-complete selection
assorted (SM)	a shape-flavored <code>factor_local_breaking</code> predicate; a minimality “currency” rescue; an $SU(2)$ cubic-anomaly bug	each detected in review; each would have faked a result	the reconstructed carrier; the alias-inflated 11,990-row denominator and its exclusions are discarded

C.3 Rigging toward a verdict, in both directions: the P1 sequence

The entanglement-kernel question (Section 6.1) passed through four constructions before the exact program; the fourth was accepted for a time and then overturned by exact quotienting.

1. **Rigged toward the positive.** A bespoke graph, with edge roles literally named `active_bottleneck` and `hidden_half`, had three parallel channels: the “invisible mode” was engineered into the carrier, and the “bulk invariant” that moved was a single edge weight relabeled as a distance. *Defect: manufactured witness.*
2. **Rigged toward the negative.** On the honest carrier, the dimension of genuine modes was hardcoded to zero (a dead conditional over an empty basis), and a scoring filter (“only multi-edge geodesics count; volume is slack”) was chosen so that the computed kernel could not count. *Defect: verdict by filter.*
3. **Inflated by degeneracy.** At the uniform-weight point (all 14 edges equal, all cuts tied), a one-sided forward-difference Jacobian reported nullity 6. Every “kernel” direction changed the fingerprint at first order, with residuals 2.3×10^{-1} , 2.3×10^{-3} , 2.3×10^{-5} at steps 10^{-1} , 10^{-3} , 10^{-5} , and single edges were invisible in one direction only (cut slack). *Defect: linearization artifact at a non-generic point.*
4. **Accepted, then overturned.** Generic tie-broken geometries, two-sided central differences, both-direction finite-range invisibility, and a tree control gave nullity 5/5/5. Exact series-parallel quotienting then showed that all five directions were reparametrizations. The exact five-class search leaves 13 explicitly unclassified residual carriers, which are the starting point of the program, not a physical prediction.

C.4 Review record

The review history has three parts. None of them is journal peer review or independent replication.

An earlier external gate. A missing-layer gate run with an external reviewer recorded “approve with changes” (round v7), then “still needs work” (v8), and then no answer (v9). No final approval was issued. The missing-layer atlas therefore cannot serve as independent preregistration or blinding evidence; its cards are a historical research record and do not override the claims registry.

The 2026 verification campaign. Twenty-seven adversarial review rounds and twenty repair or probe constructions, completed on 2026-08-26, inspected every load-bearing claim and required repairs to construct missing objects rather than relabel outcomes. One agent reproduced each review finding, a separate read-only reviewer assessed the repairs, and implementation was commit-gated. A further model-based adversarial round, with its own reruns and audit scripts, examined the construction programs; its two central findings were reproduced and accepted. First, forward reachability had been mistyped as symmetric-orbit closure; the result is now stated as forward-reachability disjointness only. Second, the review constructed the diagonal intertwiner that makes the six state-level pairs gauge-equivalent, which led to the general gauge-collapse lemma. The campaign produced strengthened results (the conjugation-consistent branch carrier, exact $SU(5)$ generator action, the unique content orbit, genuine generic mass ranks, field provenance, and explicit min-cut duality) as well as decisive negative findings (the commuting routes, the reparametrization kernel, the token-sensitive record implication, the failed linear response, the unshared Born/area composition, and the solver-artifact F50 boundary). Its dispositions (keep, restate, retract, open program) are recorded in review_2026/CLAIMS_MAP.md and review_2026/CLAIMS_REGISTRY.md.

The mathematics review. A subsequent review, recorded in review_2026/mathematics_audit_20261003/, re-examined all 42 registry records for quantifiers, hypotheses, and the strength of their support, and repaired what it found: it supplied the general proof of Theorem 7.1 with exact rank certificates and an exact partial-readout witness, extended the gauge-collapse certificate to all connected carriers, strengthened several validators, and narrowed the record census to candidate channels. It is a self-review with a separate adversarial pass.

C.5 What the trail shows

Three things. First, the checks caught failures in both directions, though not always immediately. Second, a passing validator is necessary but never sufficient: semantic defects survived green validators. Third, the paper accordingly claims no surviving forced predictions. It reports finite constructions and open programs, and it keeps the specific failed definitions and controls as regression tests.

D Notation and key numbers

D.1 Recurring symbols

symbol	meaning	first use
H, q, Q, t	carrier, access map, layer, readout	§2.1
$\mathcal{O}_q(t)$	obstruction set: witnesses that t does not descend through q	§2.1
$\Delta_{\text{fact}}(\pi_0, \pi_1)$	factorization defect: witnesses that π_1 is not a function of π_0	§2.1
$P_1, \dots, P_6$	the six role names (descent, representability, route mismatch, refinement, packaging, audit)	§2.2

symbol	meaning	first use
F_n	structural law n of the catalog (Appendix A.5)	§2.2
$2 3$	gauge structures whose nonabelian factors are $SU(2)$ and $SU(3)$	§4.3
$\Delta_3^{\text{comp}}(E_1, E_2)$	route-mismatch defect of two idempotent completions	§5.1
$L = (d_0, d_1, d_2, d_3)$	the QM-GR common refinement; $q_{\text{QM}} = (d_0, d_1, d_2)$, $q_{\text{GR}} = (d_0, d_2, d_3)$	§5.2
ψ, V_0	field and background potential of the 54-record construction	§5.2
$T_{\text{QG-NoGo}}$	narrow statement about two fused or stapled routes	§5.2
$S(A), \text{mincut}(A)$	entanglement entropy; geometric minimum cut (“area”)	§5.4
c_e, y_e	edge capacity; dual variable, the per-edge shadow price	§5.4
$I_3(A:B:C)$	tripartite information; monogamy means $I_3 \leq 0$	§5.4
$\text{RS}(C, \phi)$	record-stability predicate on a branch	§6.3
C2_L1	the binary-copy tensor family of the P1 state-level route	§6.1
$M = LR^\top, G$	raw boundary matrix of the toy holographic carrier; internal gauge in $GL(d)$	§7.2
$V[\psi] = V_0 + \kappa T_{00}[\psi]$	semiclassical fixed-point map of the F50 carrier	§7.3

D.2 Key numbers

quantity	value and conditions
gauge census	1,066 $\rightarrow$ 52 labelled; 419 $\rightarrow$ 24 declared orbits; 195 $\rightarrow$ 24 primitive-charge orbits
branch selection	clean branches exist only for $2 3$, all $SU(2)$ -active; every $2 3$ structure also has a breaking branch
embedding ratios	$\sin^2 \theta_W = 3/8$, $k_Y = 5/3$; regular embedding, $\bar{\mathbf{5}} + \mathbf{10}$, weak-pair normalization; Pati-Salam also gives $3/8$
mass ranks	charged 14, 28, 42, 56 and neutral deficiencies 1, 2, 3, 4 for $N = 1, \dots, 4$
F47 naturalness	declared cell 2.247%; declared sweep 0 to 40.6%
record census	on the 12 singleton branches at threshold $\theta = 2$ (and at $\theta = 1, \dots, 4$), the complete bounded candidate census refutes all three readings, with or without scalar dressing; 556 candidate lifts give 404 labels under a declared VEV-insertion identification
coordinate defects	8 unordered witness pairs in each direction; nested control 0; 55/120 incomparable partition pairs
route mismatch	completions commute; 0.5 is an encoding-dependent table distance
min-cut saturation ($D = 2, 3, 4$)	mean $S/\text{mincut} = 0.729, 0.904, 0.941$ over five seeds (sampled)
monogamy	min-cut $I_3 \leq 0$; GHZ $I_3 = +\log 2$
CHSH	quantum $2\sqrt{2}$; enumerated local bound 2; classical control $\sqrt{2}$
information loss	exact rank 27 for both 81×27 frames of each of three seeds; half-boundary witness $\langle Z \rangle = \mp 4.674 \times 10^{-3}$
F50 sweep	fixed points for every sampled background and κ ; raw-map stability crosses near offset 3.6535, $\kappa = 5.8675$
P1, graph level	19 exact pairs on 13 carriers; forward-reachable sets disjoint; symmetric closure open
P1, state level	six exact joint-map fibers; all gauge-equivalent by the C2_L1 lemma

quantity	value and conditions
P2, BMV	an assumed perfect-record channel is LOCC and entanglement-null; the channel is not derived

E Version history

Version 1 (17 June 2026). Initial release.

Version 2 (31 August 2026). A systematic verification of Version 1 (Appendix C.4). The following Version 1 statements are not retained as claims: route non-commutation for the constructed completions and an explanation of why quantizing gravity fails; universal or adversarial uniqueness of the QM-GR parent; a computational grounding of the common carrier; a discrete linearized-Einstein match; a universal shared Born-area composition law; an unconditional 99.3% exclusion over an 11,990-row carrier; clean selection of a bare structure; record-stability grounding; the N_{gen} -blindness theorem and the content-blindness comparison; the F24 architecture verdict; a stable 3.96% naturalness fraction; the product-parent 3/23 control; the three forcing predictions; the F50 15/18 split and κ existence boundary; E032 as a theorem extending quantum mechanics; and the description of the external missing-layer gate as settled. The λ “adjudication” of the E021 card is archived exploration, since its contraction target and measure center are imported. The claim-by-claim mapping is in review_2026/CLAIMS_MAP.md.

Version 3 (3 October 2026). A mathematics review of all 42 registry records, and a rewrite of the text for readability with new figures. Changes of substance:

- the information-loss result now rests on a general factorization theorem (Theorem 7.1) with exact rank certificates and an exact half-boundary witness, replacing sampled reconstructions and a floating-point witness; the raw-matrix and normalized-ray readings are distinguished;
- the gauge-collapse lemma is certified on every nonempty connected carrier, including those with zero or one edge, and disconnected carriers are rejected;
- the step-61, step-27, step-28, and step-69 validators were strengthened (full rebuilds, reconstruction from tracked inputs, and a literal source pin), and an independent representation-theory oracle was added;
- the record census is described as a census of candidate channels, whose counts bound but do not compute the rank of a restriction map, and the statement that excluding scalar-dressed candidates restores the implication was corrected (on the 12 singleton branches at the reference threshold it fails with and without dressing; among the tested definitions only the restricted generator grammar, at thresholds $\theta \geq 2$, restores it);
- imported statements are scoped precisely: the no-algebra theorem concerns decoding of judgment status; F34 recovery is on the readout’s image; F37 is relative to a declared admissible class; F49 includes a declared direct route and is not a probabilistic Bell theorem; F23, F24, F27, F39, and F48 carry their supplied premises.

No main claim was downgraded.

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